Off-Grid Inverter

KR-L12K-SP-BP

with Breakers



KR-L12K-SP-BP

12 kW OFF-GRID

120/240V SPLIT PHASE

BUILT-IN BREAKERS

User Manual

READ THE INSTRUCTIONS COMPLETELY BEFORE OPERATING THE EQUIPMENT

Check the utility voltage before turning ON the unit.

Verify the inverter's programmed grid type before connecting to the utility.

Disregarding these instructions could result in permanent damage to the unit.



DISCLAIMER

UNLESS SPECIFICALLY AGREED TO IN WRITING:

KOOR assumes no responsibility or liability for any damages, property loss, personal injury, or any adverse consequences resulting from improper use and installation of the product or the failure to adhere to the guidelines provided in this document. Users are expressly advised to follow the instructions and guidelines outlined in the documentation accompanying the product. KOOR shall not be liable for any damages or losses incurred due to deviations from recommended usage, installation, or maintenance procedures. By using the product, users acknowledge their understanding of these disclaimers and agree to use the product at their own risk. KOOR reserves the right to update or modify product information, specifications, and guidelines without prior notice.

KOOR retains the right to final interpretation of this document and all related materials pertaining to this product. This document is subject to modifications, updates, revisions, or termination without prior notice.

Fantastech Energy Solution LLC

Email: derek@fantastech.org | Phone: 574-218-1973 | Goshen, IN 46528, US

12 kW Split Phase Off-Grid Inverter with Breakers — Quick Start

KR-L12K-SP-BP | 120/240 V split phase, 60 Hz | off-grid only

Keep this sheet with the inverter. It covers the steps you use on the job. Full detail, every setting and the whole fault list are in the User Manual in this same book. Qualified electrical personnel only.

1 WHAT IS ON THE UNIT

PORTS AND PARTS

1	PV Switch
2	Power Switch
3	PV Input Terminal
4	Smart Load L1/L2 Breaker
5	AC Input L1/L2 Terminal
6	Load L1/L2 Breaker
7	PE Terminal (ground)
8	Neutral
9	Generator Input Breaker
10	INV Input Breaker
11	Communication Interface
12	BAT Terminal
13	BAT Breaker
14	Wi-Fi Port
15	LED Display
16	Run Indicator

INDICATOR LIGHTS

1	Output indicator (green) - AC OUT	Flashes when the inverter is supplying power.
2	Charging indicator (yellow) - CHARGE	Flashes during charging.
3	Fault indicator (red) - FAULT	Lights up when a fault occurs. It either stays on steady or flashes. This shows how serious the fault is.

SCREEN AND CONTROLS

Power Switch	Main Power / Emergency Output Disconnect (EPO) Switch. Closing (turning on) this switch restores normal operation and power output. In an emergency, opening (turning off) this switch immediately cuts all outputs and stops the power supply.
PV Switch (PV Disconnect)	Safely connects and disconnects the PV array from the inverter. If there is a problem with the PV modules, open the switch to de-energize the DC input circuit.
Screen wake	Press any button on the screen to wake the screen from sleep mode.
Tap Setting	Tap Setting on the main screen to open the settings page. You will see six main setting groups: Quick, EMS, Battery, Grid, Others and Advanced.
Tap any icon	Tap an icon to open that part's details page and see all of its values.
Tap the alarm code	Tap the alarm code to see the exact alarm type.

BUILT-IN BREAKERS

Battery Breaker (Battery Switch Port)	Rated Current: 300A
INV Input Breaker (INV Input Switch Port)	Rated Current: 60A
Generator Breaker (Generator Input Switch Port)	Rated Current: 60A
Load Breaker (Load Switch Port)	Rated Current: 125A (grid-powered)
Smart Load Breaker (Smart Port)	Rated Current: 60A
PV Switch / PV Disconnect	PV array disconnect switch - no current rating printed in the manual
Bypass circuit breaker phase specification (Section 11)	2-pole, 200A/200A (L1/L2)
AC charging circuit breaker phase specifications (Section 11)	2-pole, 63A/63A (L1/L2)
Generator mode circuit breaker phase specifications (Section 11)	2-pole, 63A/63A (L1/L2)

2 WHERE IT CAN HANG

Max ambient temp	55 C (131 F). Operating temperature range -10 C to 55 C (14 F to 131 F), with derating above 45 C (113 F).
Air around the inverter	Keep it between -10 C and 55 C (14 F and 131 F).
Storage temperature range	-25 C ~ 60 C (-13 F ~ 140 F)
Humidity range	5% to 95% (conformal coating protection)
Clearance above and below	200 mm (7.87 in)
Clearance to each side	500 mm (19.69 in)
Mounting angle	Allowed: tilted at >75 degrees, or straight up at 90 degrees. Not allowed: lying flat, tilted past 90 degrees, or straight up at 90 degrees but upside down.
Size (L*W*D)	760mm*485mm*160.8mm (about 29.92 in x 19.09 in x 6.33 in)
Net weight	35.5 kg (about 78.3 lb)
Noise	60dB or less
Cooling	Forced air cooling, variable-speed fan

YES

- Install the inverter in a cool, shaded place that is protected from rain.
- Mount it on a surface that can hold its weight.
- Leave enough space around the unit for heat to escape. Keep at least the clearances listed.
- Keep the mounting angle in the allowed range: tilted at >75 degrees, or straight up at 90 degrees.
- The openings on the bottom of the case are sized to industry standards - bring conduit in the bottom.

NO

- Do not install the inverter where there are flammable building materials or where an explosion could happen.
- Do not install the inverter in a harsh location, such as a humid, oily, flammable, explosive, or dusty area.
- Keep it out of direct sunlight, rain, and snow. These can damage the inverter or shorten its service life.
- Do not lay it flat, tilt it past 90 degrees, or mount it upside down.
- Do not connect one PV array to two inverters.

3 MOUNTING

1. Step 1: Hold the drill template on the wall. Mark the holes, then drill them.
2. Step 2: Install the expansion bolts and the wall-mount bracket. Check the bracket orientation (arrow points up).
3. Step 3: Lift the inverter and hang it on the wall mount.
4. Step 4: Secure the inverter with the right-angle brackets (supplied) on both bottom sides.
5. Step 5: Tighten the right-angle bracket. Repeat on the other side. The installation is finished.

4 WHAT IS IN THE BOX

Parallel cable	1
Hex key (2.5 mm)	1
Hex key (8 mm)	1
Wall-mount bracket	1
Wall-mount (right-angle) bracket	2
Ring terminal (M8)	4
Anchor bolt	4 (listed twice in the packing list)
Mounting hole template	1
Product manual	1
Hex flange nut (M8)	5
Expansion tubes	2
Self-tapping screws	2

Check the box and the unit for damage before you sign for it. Note any dents, holes or cracks on the freight paperwork.

5 CABLES

Battery	Breaker type 2P-300 A; rated discharge current 270 A	53.5 mm2 (1/0 AWG) x2P
AC input (grid)	Breaker type 3P-200 A; max bypass input current 200 A	42.4 mm2 (1 AWG) x2P
GEN input	Breaker type 3P-63 A; maximum GEN input current 60 A	13.3 mm2 (6 AWG)
LOAD output	Breaker type 3P-125 A; maximum load output current 125 A	33.62 mm2 (2 AWG)
Smart Load output	Breaker type 3P-63 A; maximum load output current 60 A	13.3 mm2 (6 AWG)
PV input	Breaker type 2P-40 A; maximum PV input current 32 A	5.2 mm2 (10 AWG)
Battery wiring entry	Conduit opening 1.98 in. (50.4 mm)	Recommended conduit 1 1/2 in.
COM wiring entry	Conduit opening 1.37 in. (34.9 mm)	Recommended conduit 1 in.
GEN wiring entry	Conduit opening 1.98 in. (50.4 mm)	Recommended conduit 1 1/2 in.
Load wiring entry	Conduit opening 1.98 in. (50.4 mm)	Recommended conduit 1 1/2 in.
GRID wiring entry	Conduit opening 1.98 in. (50.4 mm)	Recommended conduit 1 1/2 in.
PV wiring entry (PV1, PV2)	Conduit opening 1.37 in. (34.9 mm)	Recommended conduit 1 in.

TORQUE

Battery terminals	2-3 Nm (17.7-26.6 lbf-in)
AC input (grid) terminals	1.2-1.6 Nm (10.6-14.2 lbf-in)
GEN input terminals	1.2-1.6 Nm (10.6-14.2 lbf-in)
LOAD terminals	1.2-1.6 Nm (10.6-14.2 lbf-in)
Smart Load terminals	1.2-1.6 Nm (10.6-14.2 lbf-in)
PV terminals	1.2-1.6 Nm (10.6-14.2 lbf-in)

BEFORE YOU TOUCH A TERMINAL

- Only certified electricians may install or wire the inverter. Turn off (de-energize) all power sources first.
- Turn OFF the power switch before you start any wiring.
- The inverter must be properly grounded. When wiring, connect the ground wire first. When disconnecting, remove the ground wire last.
- Always connect the protective earth (PE) wire first.
- Check the polarity markings closely so the circuit polarity is correct. Never reverse the battery polarity.
- In parallel systems, you must get the polarity right during installation. Wrong polarity can cause a severe short circuit.
- Remove the terminal cover before you start wiring. Put it back on after all connections are made.
- Install a DC protection device between the battery and the inverter, and a protective switching device between the utility grid and the inverter.
- Install a lightning protection device (LPD) on the inverter's DC input side.
- Do not take the inverter apart while it is running.
- Danger of electric shock! Before any maintenance work, disconnect every power source to the unit and make sure all capacitors have discharged.
- Turn the inverter OFF and switch the battery off before you plug in or unplug a communication cable.

6 PV CHECK BEFORE YOU LAND IT

Max PV array open circuit voltage	500Vdc (maximum open-circuit voltage of each MPPT channel is 500Vdc)
MPPT voltage range	90Vdc~450Vdc
PV operating voltage range	90Vdc~500Vdc
Max PV input current	32A + 32A (each MPPT channel supports a maximum operating current of 32A)
Max PV input power	12000W + 12000W
Number of MPPT groups	2
Strings per channel	Each channel has two input ports, so two PV strings can be connected in parallel.
Per-channel output	Each MPPT channel supports up to 12kW output, for a total system capacity of 24kW.
PV charge current (can be set)	250A
Voltage margin	Open-circuit voltage changes with the seasons and temperature, so leave enough voltage margin that it never goes over the limit during operation.
Parallel systems	On every inverter, each PV input must connect to its own independent, separate PV (solar) array.

Meter every string open circuit before you land it, and correct for the coldest day of the year. Voc must stay under the maximum PV array open circuit voltage above.

7 GENERATOR

Rated input voltage	120(L1/N, L2/N)/240Vac(L1/L2)
Input phase voltage range	(85Vac~140Vac) +/-2%
Frequency range	40+/-0.3Hz~70+/-0.3Hz
Maximum GEN input current	60A (adjustable from 0 to 60A)
Max Gen Input Current setting	60A default. Range: 0-60A.
Battery charging current from generator	Adjustable from 0 to 160A
Maximum charge current (can be set)	160A
Generator THD	Should stay below 3% when the generator is connected directly to the generator port
Generator sizing	For the best THD while meeting both load and battery charging needs, the generator should be at least 1.5 times the inverter's output power.
Solution 1	A 15kW or larger generator is recommended as backup power during grid outages.
Solution 2	Hybrid operation, where a smaller generator works together with the battery system (set up in the LCD menu). The generator does not need to be larger than the inverter's rated power.
Auto start/stop	Controlled by EMS Setting > Inv Switch Param: Bat to Mains Volt / Mains to Bat Volt / Bat to Mains SOC (BMS enabled) / Mains to Bat SOC (BMS enabled).
Dry contact start/stop	Starting state: NC-N is closed and NO-N is open. When battery voltage falls below the grid-switching threshold, NC-N opens and NO-N closes.
NO-N contact rating	Resistive loads rated 125VAC/1A, 230VAC/1A, or 30VDC/1A
Circuit breaker phase specifications	2-pole, 63A/63A (L1/L2)

KOOR recommends: where the manual gives both a minimum generator size and 1.5 times the inverter output, size the generator on the larger of the two.

8 COMMUNICATION PORTS

Parallel_1 (PARA_1)	RJ45 parallel communication port. Connect only PARA_1 to PARA_2 between inverters.
Parallel_2 (PARA_2)	RJ45 parallel communication port. Never connect two ports of the same type.
BMS Port	Battery BMS communication. In a parallel system, connect the battery BMS to the master inverter.
DRY	Dry contact port: IP-DRY, GND, C, NO, NC. Generator start/stop and remote AC output control (IP-GND connected = AC output OFF).
RSD_12V Power	12V DC supply for the rapid shutdown system.
RSD Input Control/BAT_T	RSD terminal: ES+, ES-, +, BAT_T. The 12V output shuts off when the ES button closes or 12V DC is applied to + and -.
CT Port	CT sampling detection (optional). Watches inverter output current to stop reverse power flow into the grid.
USB Port	USB Type B, for the optional PC host software. The computer must have the USB to Serial Port Chip CH340T Driver installed.
Wi-Fi Port	Optional Wi-Fi/GPRS module that converts the 485 signal. Installs through the RJ45 interface (RJ45 and USB are compatible - choose one) or the aviation plug interface.

BATTERY BMS PORT PINOUT

1	485-B
2	485-A
3	-
4	CAN_H
5	CAN_L
6	-
7	485-A
8	485-B

9 TURN IT ON — IN THIS ORDER

1. Make sure all installation and wiring are correct before you turn the power on.
2. Make sure the inverter is fully assembled before you turn it on. Finish all wiring, then put the terminal cover back on and tighten the mounting screws securely.
3. Close (turn on) the Battery Breaker first, before you activate the battery BMS. This order helps prevent damage to the switch from inrush current surges.
4. Press the Power Switch to turn the inverter on.
5. Keep the INV Input Breaker closed (on) during normal operation. If it is opened, grid-to-battery charging stops.
6. Set the battery, grid and advanced settings from the LCD: tap Setting, then Quick, EMS, Battery, Grid, Others or Advanced.
7. Set the output phase while the inverter is in standby mode. Make sure the inverter wiring is set up correctly.
8. Parallel systems: set each inverter one at a time with all the other inverters powered off, then restart the whole system.
9. When the system is running, measure the output voltage and confirm it is correct. Then connect the loads.

10 SHUT IT DOWN

1. In an emergency, open (turn off) the Power Switch. This immediately cuts all outputs and stops the power supply.
2. Turn OFF the power switch before you start any wiring.
3. Turn off the switch before you connect battery wires.
4. Turn the inverter OFF and switch the battery off before you plug in or unplug a communication cable.
5. If you add or remove an inverter in a running parallel system, turn off all power switches before you do any wiring.
6. Before any maintenance, disconnect every power source to the unit and make sure all capacitors have discharged. Only then check or work on the unit.
7. If there is a problem with the PV modules (fault, maintenance, or fire risk), open the PV switch to de-energize the DC input circuit.
8. Remote off: connect IP-GND on the dry contact port to turn the AC output OFF.

11 KOOR BATTERY COMMUNICATION

1. Turn the inverter OFF and switch the battery off before you plug in or unplug a communication cable. Set the address switches with the battery off.
2. Pick one battery as the master. Set its address switches all OFF (address 0). Up is ON.
3. Run a plain straight ethernet cable (CAT5e or CAT6) from the master's CAN jack to the inverter's battery communication (BMS) port. Only one battery talks to the inverter.
4. Active Stack and 16K Pro wire the same. The CAN jack is the right-hand jack of the RS485-1/CAN pair. Pin 4 CAN-H, pin 5 CAN-L on both ends. No crossed or custom cable.
5. Chain the rest in a line on RS485-2: out of B on one battery, into A on the next. The master's A jack and the last battery's B jack stay empty. RS232 and DRY are not used.
6. Give every other battery its own address: 1, 2, 3 and up to 14. Never run a cable from the last one back to the master. Changed a switch with the battery on? Hold RST 3 s, then 3 s again.
7. Set Pylon on the battery too: every pack's CAN protocol = Pylon (Active Stack: battery app, password 1234), then RST 3 s, then 3 s again.
8. Set Battery type to LFP16.
9. Set BMS Setting to BMS-CAN.
10. Set BMS Protocol Select to Pylon.
11. Set Charge Logic Select to BMS, then save.
12. Backup path only: if there is still no battery data, move the cable to the battery's RS485-1 jack, set BMS Setting to BMS-485 and BMS Protocol Select to Pylon.

> Full battery detail — port, address switches, settings and what to do when it will not talk — is in Appendix A of the User Manual in this book.

12 POST-INSTALL CHECK

- ☐ Put the terminal cover back on and tighten the mounting screws securely.
- ☐ Measure the output voltage and confirm it is correct. Then connect the loads.
- ☐ Check that the indicators and display match how the device is operating.
- ☐ The screen shows the battery state of charge as a percent from the BMS, with battery voltage and current.
- ☐ No BMS fault code is showing.
- ☐ Charge current follows the battery's own limit instead of the value you set by hand.
- ☐ Watch for any faults or errors on the display, and fix the cause if needed.
- ☐ Check all wiring terminals and tighten the screws.
- ☐ Make sure nothing blocks the airflow around the unit.

13 TOP FAULTS

Code	Fault	First thing to check
/	No display on the screen	Check that the battery circuit breaker or the PV circuit breaker is closed, and that the switch is in the ON position. Press any button on the screen to wake the screen from sleep mode.
[05]	Battery not connected	Check if the battery is not connected, or if the battery circuit breaker is not closed.
[77]	Grid Circuit Open	Check if the AC circuit breaker is closed (on). If it is open, close the breaker and restart the inverter to clear the fault.
[49]	BMS communication error	Check that the communication cable is connected correctly. Check that BMS is set to the communication protocol that matches your lithium battery.
[01] [02]	Battery under voltage warning / protection	Charge the battery until it reaches the low voltage disconnection recovery voltage.
[06]	Battery over voltage	Measure the battery voltage to see if it is higher than rated. If it is, discharge the battery until its voltage is below the battery's overvoltage recovery point.
[18]	The PV input voltage is over voltage	Use a multimeter to check if the PV input voltage is higher than the maximum rated input voltage.
[21]	PV ISO Low	The PV insulation resistance (impedance) is low. The inverter shuts down.
[09] [10]	Bypass overload protection, inverter overload protection	1. Use fewer electrical loads. 2. Restart the unit to bring back the load output.
[11]	Inverter short circuit protection	1. Check the load wiring carefully and fix the short-circuit points. 2. Power the unit up again to bring back the load output.
[22] [23]	Heat sink over temperature protection	When the unit cools below the recovery temperature, normal charge and discharge control starts again.
[27]	Fan failure	Check if the fan is not turning or is blocked by a foreign object.
[73]	DC AFCI Abnormal	PV arc fault detected. Check the installation and wiring.
[40] [42]	Parallel connection fault	Check whether the parallel cable is connected poorly, such as a loose or wrong connection.
[37]	Parallel ID setting error	Check whether the same parallel ID number is set on more than one unit.
[39]	Inconsistent AC input source in parallel mode	Check whether the parallel AC inputs all come from the same input source.
[41]	Inconsistent system firmware version in parallel mode	Check whether every inverter has the same software version.

Full fault list, settings and wiring diagrams are in the User Manual in this book. Questions before you energize it — call KOOR support at 574-218-1973.

Limited Warranty — Terms

KOOR KR-L12K-SP-BP 12 kW Split Phase Off-Grid Inverter with Breakers · sold and warranted by Fantastech Energy Solution LLC

WHAT IS COVERED AND FOR HOW LONG

KOOR off-grid inverters — 5 years. KOOR batteries — 10 years. The term runs from the date of installation, and in no case more than six months past that term from the date the unit left the factory.

The warranty covers defects of design, components and manufacturing. It applies to units bought from Fantastech and installed at the address on the warranty card. It is non-transferable.

WHAT IS NOT COVERED

- Incorrect installation or commissioning.
- Failure to follow the User Manual, the installation guide or the maintenance instructions.
- Unauthorized modifications, changes or attempted repairs.
- Incorrect use or inappropriate operation, including reversed battery polarity and use of the wrong battery communication protocol.
- Insufficient ventilation of the device.
- Failure to observe applicable safety regulations and electrical code.
- Breaking the product seal — opening the casing — without written approval.
- Grid-tied use. These are off-grid inverters and are not certified for grid connection.
- Force majeure: lightning, over-voltage, storm, flood, fire.
- Transport damage, and superficial or cosmetic defects — dents, marks, scratches — that do not affect how the inverter works.

HOW A CLAIM WORKS

- Report it within 14 days of the first sign of a problem. Call 574-218-1973 or email derek@fantastech.org.
- Send the completed warranty card, the purchase invoice, the installation or commissioning report, and any extension warranty card if you have one.
- Send pictures and the fault code or fault screen. It is the fastest way to get a unit moving.
- Fantastech decides whether the unit is repaired on site, repaired at the shop, or exchanged for a replacement device of equivalent value for model and age.
- On an exchange, the remainder of the original warranty carries over to the replacement device. A replacement does not start a new warranty term and does not come with a new certificate.
- Because of ongoing product improvement, a replacement part or unit may not match your system monitoring or other components, and the cost of changing those out is not part of this warranty.

LIMITS

- This warranty covers the equipment. It does not cover labor, travel, lift or crane rental, lost production, lost profits, or other direct or indirect costs, unless Fantastech agrees to them in writing beforehand.
- Claims beyond the rights stated here — including costs of removal and reinstallation — are not covered.
- Your rights under statutory liability and product liability law are unaffected by this warranty.

Keep the warranty card and your invoice together. Both are needed to support a claim. If the card was never sent in, send it with the claim.

Contents

1. About This Manual **3**

1.1 Who This Manual Is For 3

1.2 Safety Instructions 3

1.3 System Overview 3

1.4 Product Features 4

2. Product Overview **5**

2.1 Functional Modules and Interfaces 5

2.2 Terminal Port Functions and Specifications 6

2.3 Bottom View and Wiring Entries (Conduit Openings) 8

3. Installation **9**

3.1 Unpacking and Inspection 9

3.2 Environmental Requirements for Installation 9

3.3 Installing the Inverter 11

3.4 System Wiring Diagram 12

3.5 Start Wiring 13

3.6 Battery Connection 13

3.7 AC Input/Output Connection 15

3.8 PV Connection 18

3.9 Final Assembly 18

4. Operation **19**

4.1 Power ON/OFF 19

4.2 Operation and Display Panel 19

5. Operating Modes **34**

5.1 Charging Mode 34

5.2 AC Output Mode 35

6. Battery Type Parameters **37**

6.1 Lead-Acid Battery 37

6.2 Lithium Battery 38

7. Communication **39**

7.1 Overview 39

7.2 USB Port 39

7.3 Wi-Fi Port 40

7.4 Rapid Shutdown System 40

7.5 Dry Contact Port 40

7.6 CT Sampling Detection (Optional Feature) 40

7.7 Parallel Communication Function 41

7.8 RSD System Introduction 41

8. Parallel Installation, Wiring, and Setup **42**
8.1 Power Wiring Precautions for Parallel Systems 42
8.2 Parallel Communication Cable Precautions 43
8.3 Parallel Connection Schematic Diagrams 43
8.4 Parallel System Setup 45
9. Fault Codes and Troubleshooting **47**
9.1 Fault Codes 47
9.2 Troubleshooting 50
10. Protection and Maintenance **51**
10.1 Built-in Protections 51
10.2 Maintenance 52
11. Technical Parameters **53**
Appendix A. Setting Up a KOOR Battery **56**
A.1 The Rules That Apply to Both 56
A.2 Address Switches (both batteries) 56
A.3 KOOR Active Stack 57
A.4 KOOR 16K Pro 57
A.5 Inverter Settings 57
A.6 How to Tell It Is Talking 58
A.7 If It Will Not Talk 58

1. About This Manual

This manual covers key safety precautions, operating guidelines, instructions for use, and maintenance requirements. Follow these rules exactly when you install, operate, and maintain the inverter. KOOR is not liable for any fault or damage caused by not following this manual.

1.1 Who This Manual Is For

Only people with the certified skills listed below, and end users, are allowed to use this equipment. End users may only operate functions that do not need these certified skills.

- Knowledge of local code and regulatory requirements.
- Complete training in electrical safety.
- Complete training in installing and commissioning (starting up) electrical equipment and systems.
- Understanding of, and compliance with, all requirements in this manual.

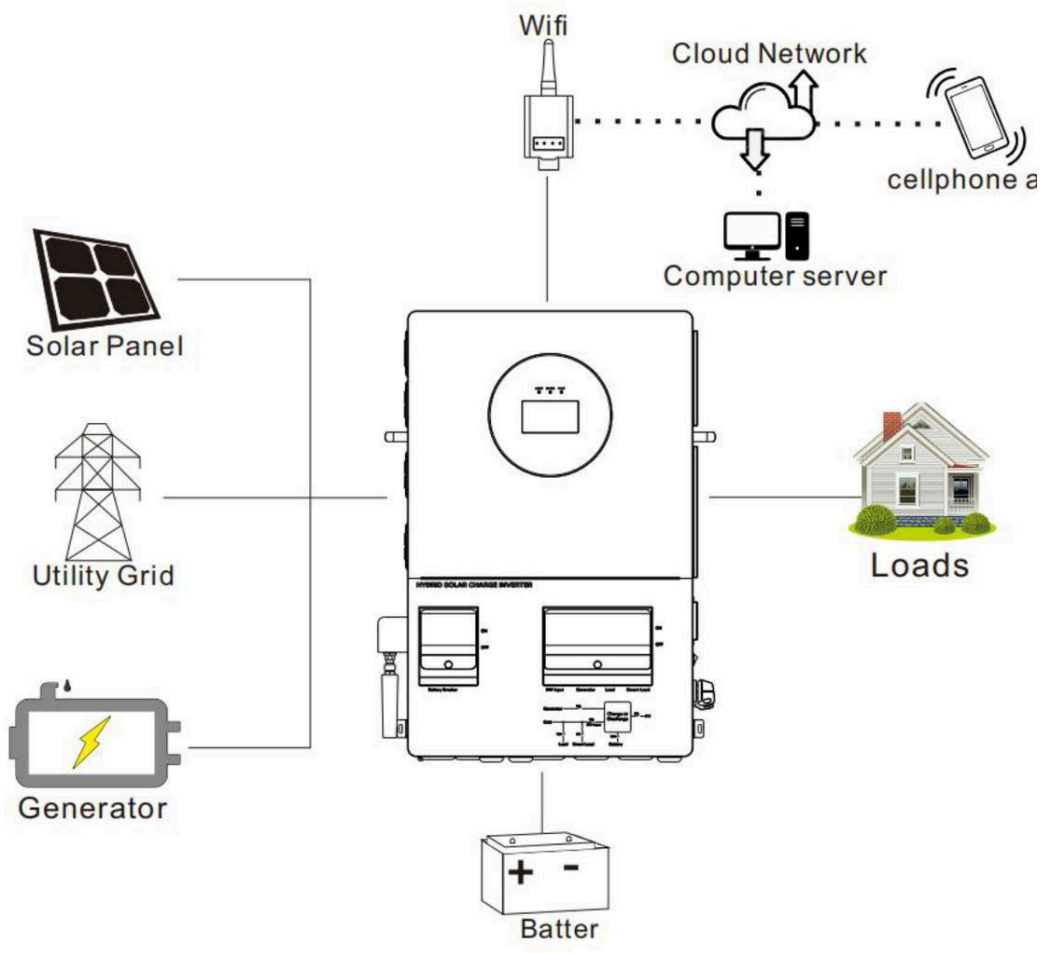
1.2 Safety Instructions

> **WARNING** Install, operate, and maintain this equipment exactly as this manual says. Any improper operation that damages the equipment or creates a safety hazard will void the KOOR warranty. KOOR is not liable for any accident that results.

- Follow local code and regulatory requirements exactly during installation.
- Only certified electricians may install or wire the inverter. Turn off (de-energize) all power sources first.
- Do not install the inverter in a harsh location, such as a humid, oily, flammable, explosive, or dusty area.
- For safe operation, choose cables that match the technical specifications. Install suitable external fuses or circuit breakers.
- The inverter must be properly grounded. When wiring, connect the ground wire first. When disconnecting, remove the ground wire last.
- Do not take the inverter apart while it is running. For maintenance or repair, send it to a professional service center.
- Choose the correct battery type. Only rechargeable deep-cycle lead-acid batteries and lithium-ion batteries are supported. Never reverse the battery polarity.
- Make sure the inverter's AC output is the only power source feeding the loads. Otherwise, damage can occur.
- Make sure the inverter is fully assembled before you turn it on.

1.3 System Overview

The diagram below shows a typical inverter system. It is for reference only. The actual wiring depends on your application. A complete system is made up of the parts shown: solar panels, utility grid, generator, battery, loads, and the Wi-Fi link (Wi-Fi dongle, cloud network, computer server, and cellphone app).



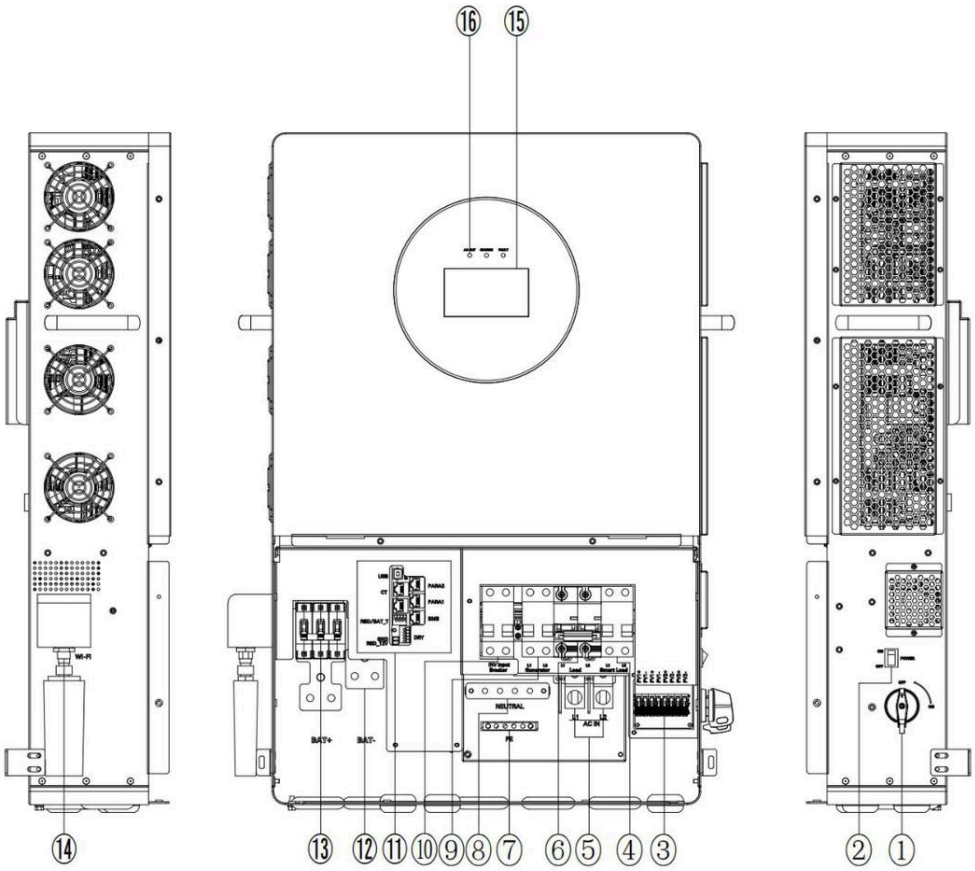
Typical system layout: solar panel, utility grid, and generator feed the inverter; the inverter powers the loads and connects to the battery; the Wi-Fi dongle links to the cloud network, computer server, and cellphone app.

1.4 Product Features

- This is an off-grid inverter. It does not support grid-tied operation (it cannot send power to the grid). It can use grid power to charge the battery while also powering the loads directly.
- Up to 6 inverters can run in parallel (72kW total output).
- The system can activate (wake up) a lithium battery using either grid power or solar power.
- Scheduled charging and discharging, with several charging modes.
- The PV (solar) system has two 12kW inputs, for a total of 24kW. Each input (circuit) accepts up to 32A.
- AFCI (arc-fault) detection. PV rapid shutdown function (RSD).
- PV insulation resistance detection (ISO).
- A dedicated generator input with dry contact control. It also has a generator efficiency enhancement function.
- Two output ports (standard load and smart load). The smart load port can be set up as an AC-coupled input.
- Communication: CAN, USB, Wi-Fi, RS485. Control and updates: local and remote.
- Full protection functions, including over-voltage/under-voltage protection, over-current and overload protection, reverse protection, and more.

2. Product Overview

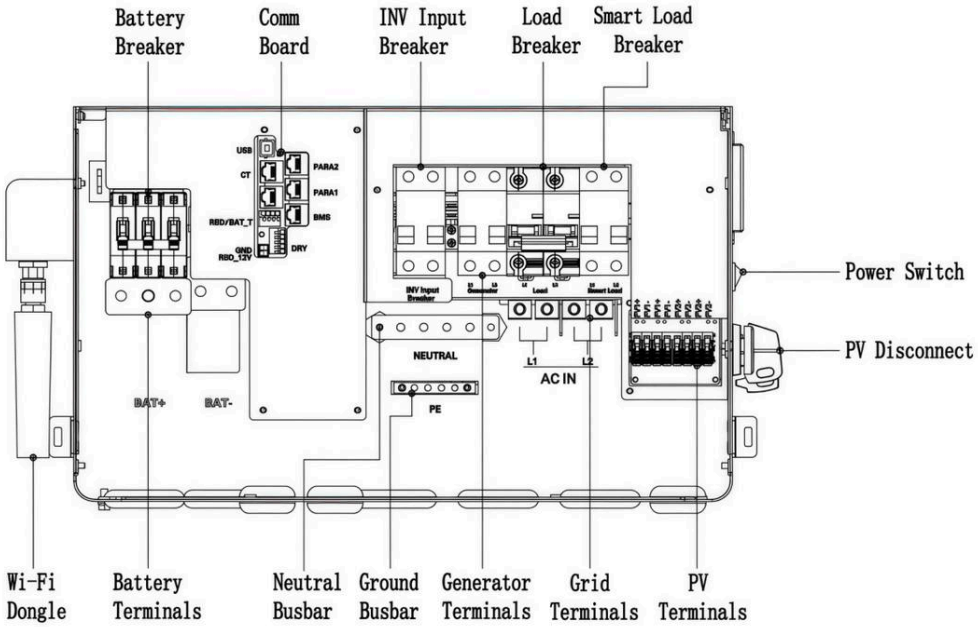
2.1 Functional Modules and Interfaces



Left side, front, and right side views with numbered parts (see table below).

No.	Part	No.	Part
①	PV Switch	⑩	INV Input Breaker
②	Power Switch	⑪	Communication Interface
③	PV Input Terminal	⑫	BAT Terminal
④	Smart Load L1/L2 Breaker	⑬	BAT Breaker
⑤	AC Input L1/L2 Terminal	⑭	Wi-Fi Port
⑥	Load L1/L2 Breaker	⑮	LED Display
⑦	PE Terminal (ground)	⑯	Run Indicator
⑧	Neutral		
⑨	Generator Input Breaker		

2.2 Terminal Port Functions and Specifications



Wiring compartment: Battery Breaker, Comm Board, INV Input Breaker, Load Breaker, Smart Load Breaker, Power Switch, PV Disconnect, Wi-Fi Dongle, Battery Terminals, Neutral Busbar, Ground Busbar, Generator Terminals, Grid Terminals, and PV Terminals.

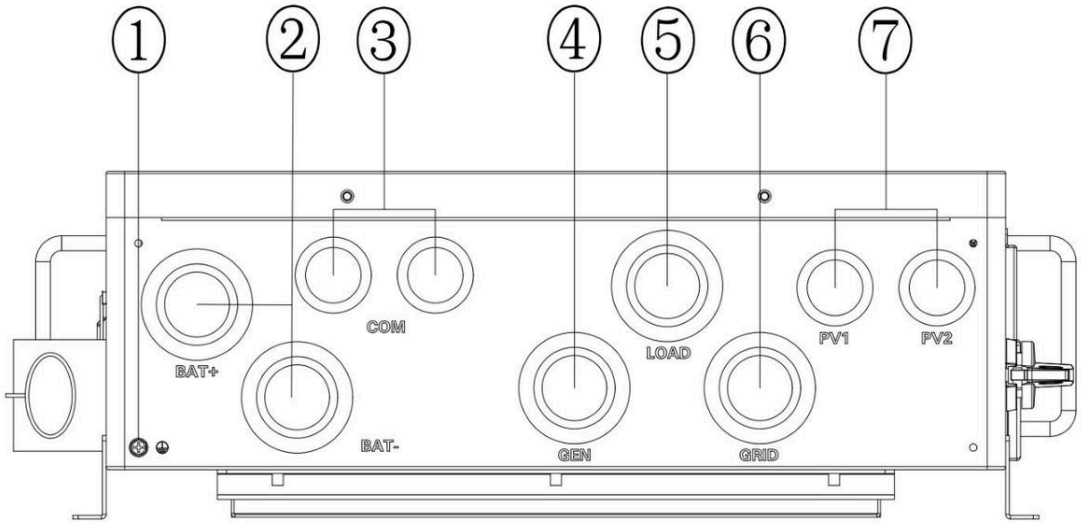
Part	Function and specifications
Battery Breaker	**Battery Switch Port (Rated Current: 300A)** **CAUTION** When starting the inverter, close (turn on) this switch first, before you activate the battery BMS. This order helps prevent damage to the switch from inrush current surges.
INV Input Breaker	**INV Input Switch Port (Rated Current: 60A)** 1. This breaker gives short-circuit protection while charging and overload protection while discharging. Keep this breaker closed (on) during normal operation. If it is opened, grid-to-battery charging stops. The breaker also gives overload protection for the load terminals in off-grid mode. 2. The system triggers **Fault Alarm 77** when it detects either of these abnormal conditions: 1. The smart load switch opens unexpectedly during grid-to-battery charging. 2. Grid power suddenly comes back while the switch is still open. The system then starts its emergency routine on its own. It tries to close the bypass relay (up to 3 attempts) to restore grid charging. If all attempts fail, do the standard emergency steps by hand: 1. Set the rocker switch (the switch marked ⑥) to OFF. 2. Close the protection switch again. 3. Set the rocker switch (the switch marked ⑥) back to ON. 4. Restart the inverter.

Part	Function and specifications
Generator Breaker	**Generator Input Switch Port (Rated Current: 60A)** 1. The system supports two generator setups. Maximum input current is 60A (adjustable from 0 to 60A). Battery charging current is adjustable from 0 to 160A. When a generator is connected directly to the generator port, its total harmonic distortion (THD) should stay below 3%. For the best THD while meeting both load and battery charging needs, the generator should be at least 1.5 times the inverter's output power. • **Solution 1:** A 15kW or larger generator is recommended as backup power during grid outages. • **Solution 2:** Hybrid operation, where a smaller generator works together with the battery system (set up in the LCD menu). The generator does not need to be larger than the inverter's rated power. 2. **Generator Auto-start Function:** The generator's start and stop are controlled by the settings under "EMS Setting > Inv Switch Param": Bat to Mains Volt / Mains to Bat Volt / Bat to Mains SOC (BMS enabled) / Mains to Bat SOC (BMS enabled). These settings are described under "EMS" in Section 4.2. For the generator start/stop wake-up wiring, see Section 7.5.
Load Breaker	**Load Switch Port (Rated Current: 125A)** The load switch is rated for 125A (grid-powered).
Smart Load Breaker	**Smart Port (Rated Current: 60A)** This port can be used as a load port or as an AC coupling port. You choose its function on the LCD screen. 1. **AC coupling:** The smart port can connect up to 60A of grid-tied solar equipment for AC coupling. That grid-tied solar equipment can then charge the battery and power the loads through the AC-coupled inverter. This inverter is a stand-alone (off-grid) inverter. It can do AC coupling, but it cannot send energy back to the grid. Pay special attention to this. The AC coupling function can be turned on and off based on battery voltage and capacity: • ON when: battery SOC/voltage ≤ set point • OFF when: battery SOC/voltage ≥ set point 2. **Smart load:** The smart load port greatly improves energy use through intelligent energy management. It lets you share or adjust the total output load capacity. Once enabled, the system can start and stop this port automatically based on battery voltage and state of charge (SOC): • ON when: battery SOC/voltage ≥ set point • OFF when: battery SOC/voltage ≤ set point
Power Switch	**Main Power / Emergency Output Disconnect (EPO) Switch** Closing (turning on) this switch restores normal operation and power output. In an emergency, opening (turning off) this switch immediately cuts all outputs and stops the power supply.

Part	Function and specifications
PV Disconnect	**PV (Photovoltaic) Array Disconnect Switch Port** 1. This switch safely connects and disconnects the PV array from the inverter. If there is a problem with the PV modules (for example a fault, maintenance, or fire risk), open the switch to de-energize the DC input circuit. 2. This inverter has a dual-channel MPPT (Maximum Power Point Tracking) system. Its parameters and operating requirements are: • The maximum open-circuit voltage of each MPPT channel is 500Vdc. When you design the PV array, its maximum open-circuit voltage must not go over this limit. Open-circuit voltage changes with the seasons and temperature, so leave enough voltage margin that it never goes over the limit during operation. • Each MPPT channel supports a maximum operating current of 32A. Each channel has two input ports, so two PV strings can be connected in parallel. 3. Each MPPT channel supports up to 12kW output, for a total system capacity of 24kW.

2.3 Bottom View and Wiring Entries (Conduit Openings)

The openings on the bottom of the case are sized to industry standards.



Bottom view with numbered wiring entries (see table below).

No.	Entry	Diameter	Recommended conduit size
①	PE Terminal (ground)	—	—
②	Battery Wiring Entry (BAT+, BAT-)	1.98 in. (50.4 mm)	1 1/2 in.
③	COM Wiring Entry	1.37 in. (34.9 mm)	1 in.
④	GEN Wiring Entry	1.98 in. (50.4 mm)	1 1/2 in.
⑤	Load Wiring Entry	1.98 in. (50.4 mm)	1 1/2 in.
⑥	GRID Wiring Entry	1.98 in. (50.4 mm)	1 1/2 in.
⑦	PV Wiring Entry (PV1, PV2)	1.37 in. (34.9 mm)	1 in.

3. Installation

3.1 Unpacking and Inspection

Check the unit before you install it. Make sure nothing inside the package is damaged.

The package should contain the items below.



Parallel cable x1



Hex key (2.5 mm) x1
Hex key (8 mm) x1



Wall-mount bracket x1



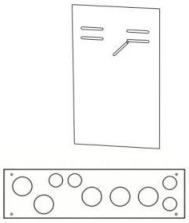
Wall-mount (right-angle) bracket x2



Ring terminal (M8) x4



Anchor bolt x4



Mounting hole template x1



Product manual x1



Hex flange nut (M8) x5



Anchor bolt x4



2 expansion tubes and 2 self-tapping screws

3.2 Environmental Requirements for Installation

Follow the points below when you choose where to install the inverter.

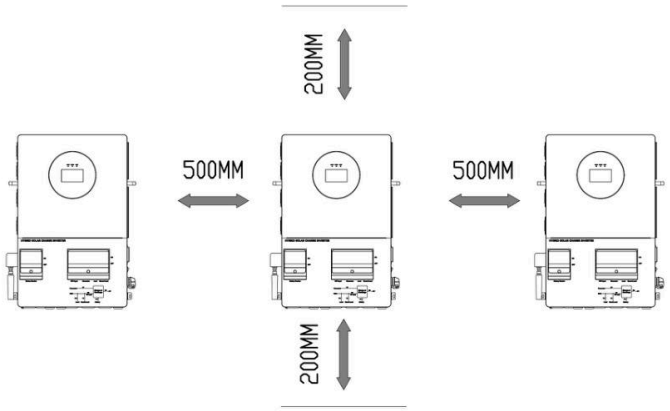
> **WARNING** Do not install the inverter where there are flammable building materials or where an explosion could happen.

CAUTION Install the inverter in a cool, shaded place that is protected from rain. Keep it out of direct sunlight, rain, and snow. These can damage the inverter or shorten its service life.



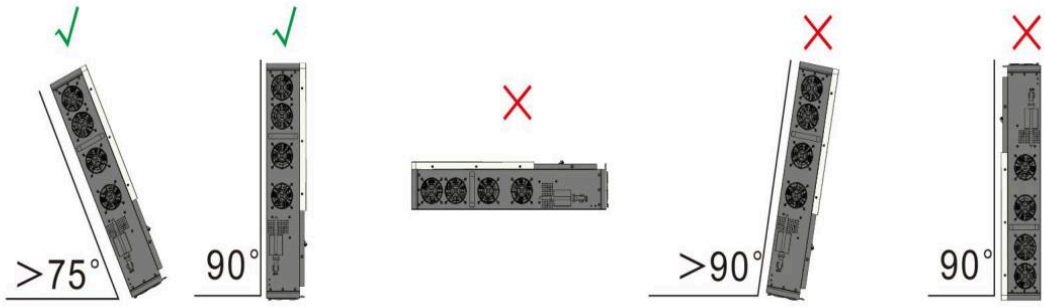
X = wrong: exposed to snow, rain, or direct sun. Check mark = right: sheltered location.

CAUTION For best operation, keep the air around the inverter between -10°C and 55°C (14°F and 131°F). Leave enough space around the unit for heat to escape. Keep at least the clearances shown below.



Minimum clearances: 200 mm (7.87 in) above and below; 500 mm (19.69 in) to each side.

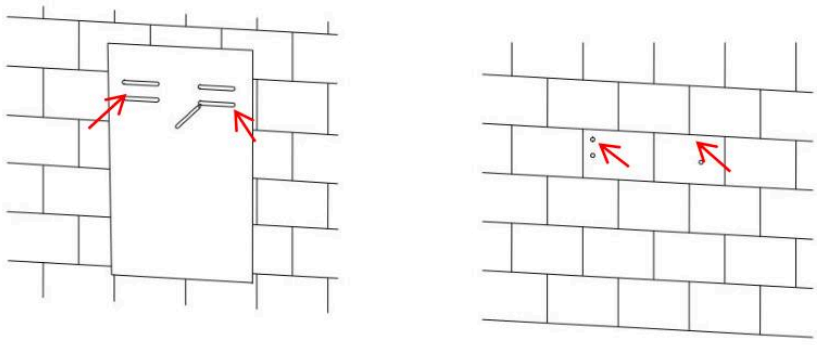
CAUTION Mount the inverter on a surface that can hold its weight. See the drawing below for the allowed mounting angle.



Allowed (check mark): tilted at >75°, or straight up at 90°. Not allowed (X): lying flat, tilted past 90° (>90°), or straight up at 90° but upside down (as shown on the far right).

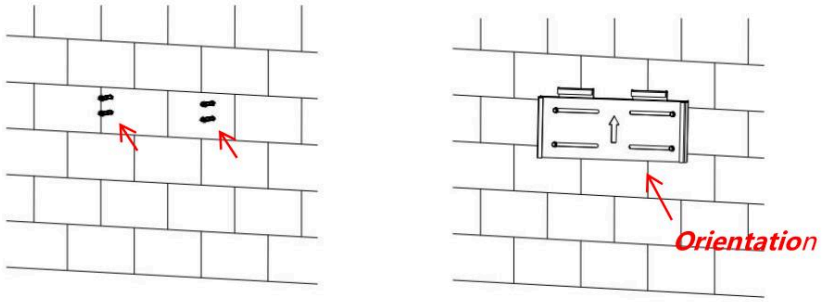
3.3 Installing the Inverter

Step 1: Hold the drill template on the wall. Mark the holes, then drill them.



Step 1: mark the holes with the template, then drill.

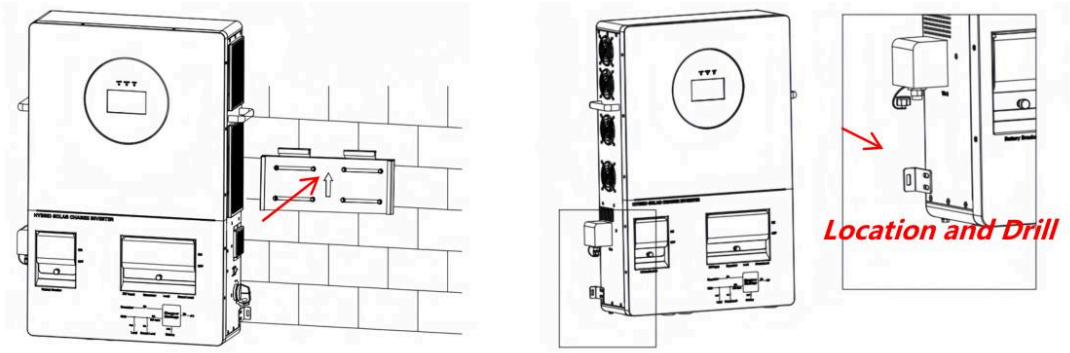
Step 2: Install the expansion bolts and the wall-mount bracket.



Step 2: install the expansion bolts, then the bracket. Check the bracket orientation (arrow points up).

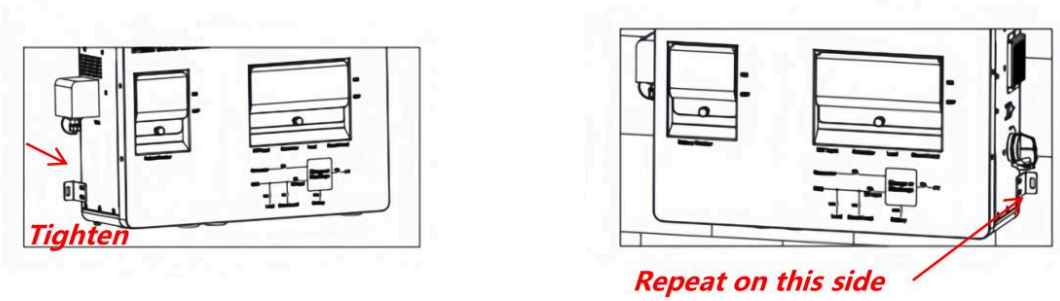
Step 3: Lift the inverter and hang it on the wall mount.

Step 4: Secure the inverter with the right-angle brackets (supplied) on both bottom sides.



Steps 3 and 4: hang the inverter on the bracket. Locate and drill for the right-angle bracket.

Step 5: The installation is finished.



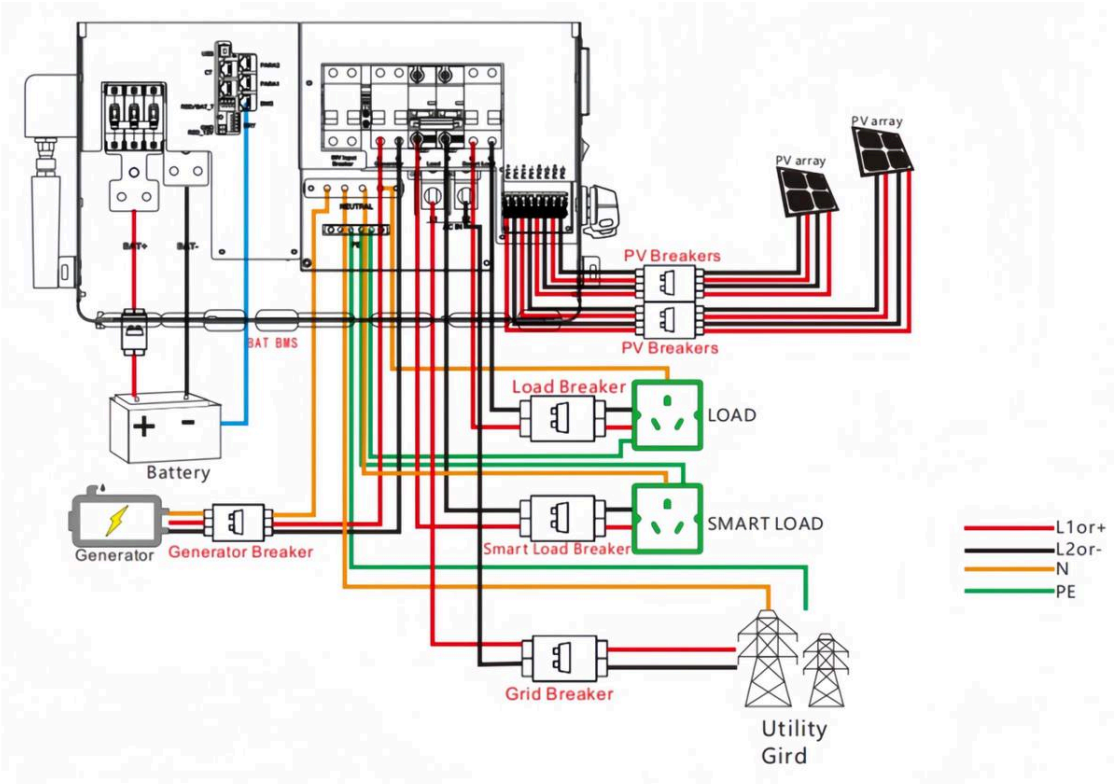
Step 5: tighten the right-angle bracket. Repeat on the other side.

3.4 System Wiring Diagram

CAUTION This wiring diagram is for reference only. The actual connections depend on your application.

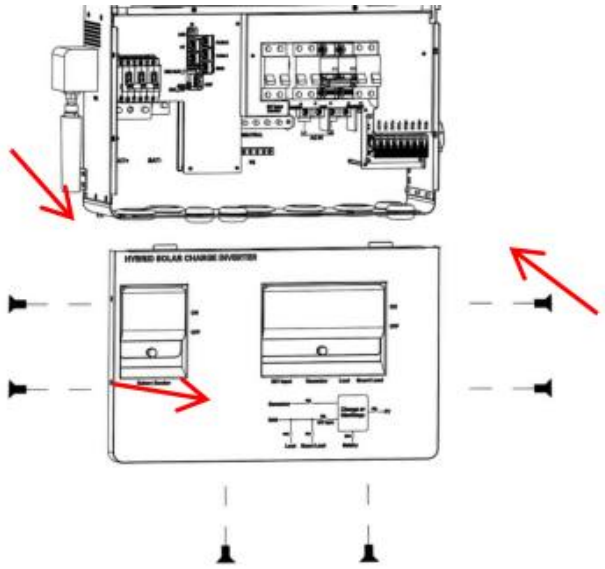
Neutral-to-ground connection: The neutral is solidly bonded between the AC input and the AC output (common-neutral design). This connection is always in place. The system needs only one neutral-to-ground connection. This is usually the main bonding jumper at the first utility disconnect. If there is no such bond, the inverter can make this connection internally through a setting.

Setting: **N Line Ground** (see "Grid (AC Setting)" in Section 4.2).



System wiring: battery (with BAT BMS cable), PV arrays through PV breakers, load through the load breaker, smart load through the smart load breaker, generator through the generator breaker, and utility grid through the grid breaker. Wire colors: red = L1 or +, black = L2 or -, orange = N, green = PE.

3.5 Start Wiring



Removing the terminal cover screws

- Remove the terminal cover before you start wiring.
- Put the terminal cover back on after all connections are made.

3.6 Battery Connection

CAUTION For safe operation and to meet code, install the equipment as follows.

- Install a DC protection device between the battery and the inverter.
- The battery cables must be rated for the required current.
- Make sure the terminals on the battery cables can carry the full rated current.

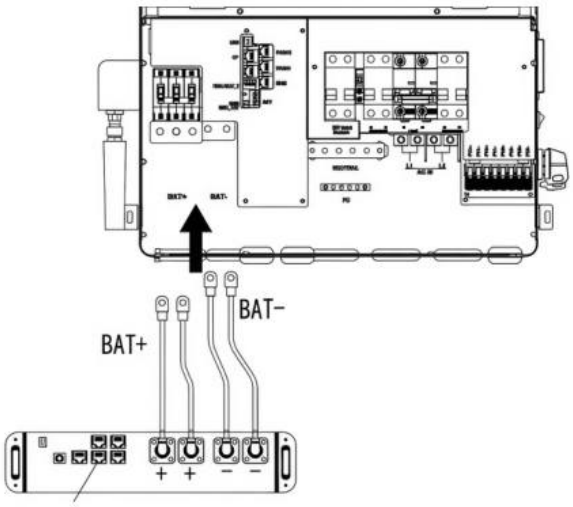
Recommended Battery Cable and Breaker Specifications

Model	Wire size	Rated discharge current	Breaker type	Torque value
KR-L12K-SP-BP	53.5 mm² (1/0 AWG) ×2P	270 A	2P—300 A	2-3 Nm (17.7-26.6 lbf·in)

3.6.1 Lead-Acid Battery

Use a 48 V lead-acid battery with enough capacity. The recommended charging current is 0.1–0.25C (C = battery capacity).

Battery Connection Diagram



Lead-acid battery connection: BAT+ and BAT-

WARNING

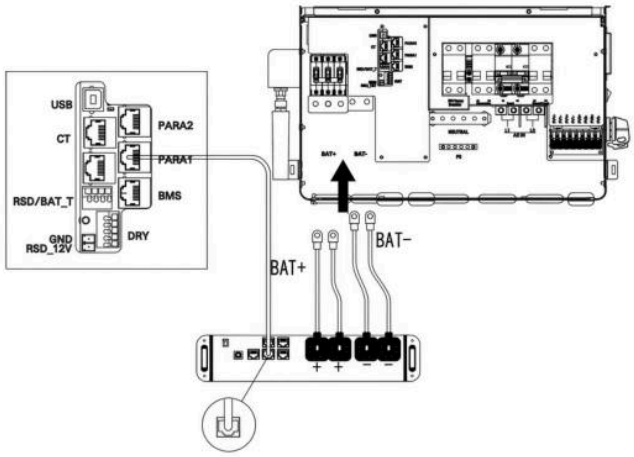
- Turn off the switch before you connect the wires.
- Make sure the battery terminal polarity is correct.
- The positive and negative cables must be the same length.
- Make sure all battery connections are tight.

3.6.2 Lithium Battery Connection

> Using a KOOR battery? **Appendix A** gives the exact port, addresses and settings for the KOOR 16K Pro and the KOOR Active Stack.

The battery and the inverter must use compatible communication protocols.

A. Battery Connection Diagram



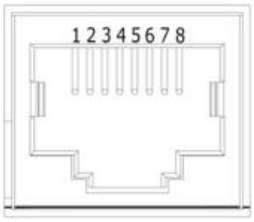
Lithium battery connection: BAT+, BAT-, and the communication cable to the inverter's communication ports

WARNING

- Turn off the switch before you connect the wires.
- Make sure the battery terminal polarity is correct.
- Keep the positive and negative battery cables the same length.
- Make sure the battery connections are secure.

CAUTION The battery and inverter need a BMS communication connection.

B. RS485 Communication Port

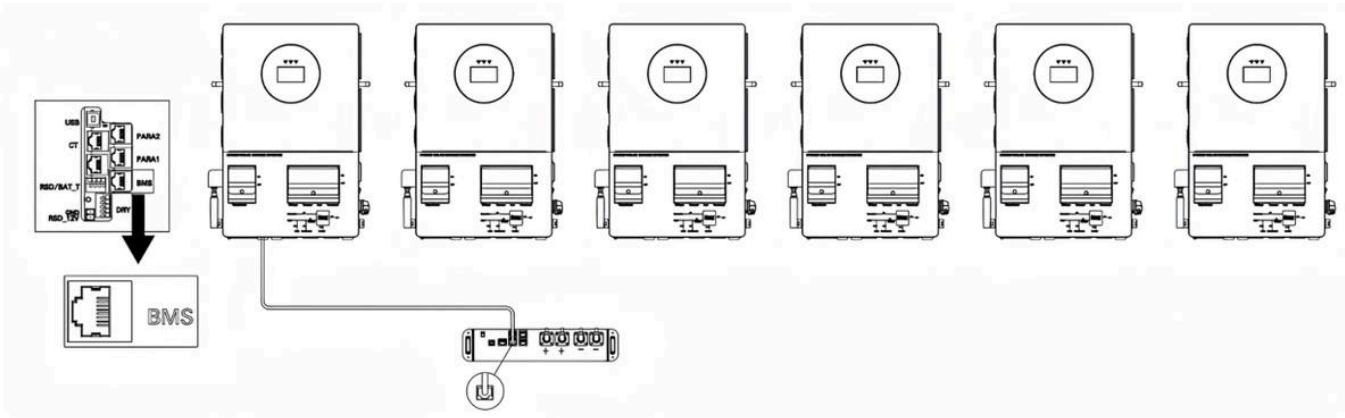


RJ45 port, pins 1–8

RJ45	Function
PIN_1	485-B
PIN_2	485-A
PIN_3	—
PIN_4	CAN_H
PIN_5	CAN_L
PIN_6	—
PIN_7	485-A
PIN_8	485-B

C. Communicating with the Battery BMS in a Parallel System

CAUTION In a parallel system, connect the battery BMS to the master inverter for best performance. (Connecting it to any slave unit also works.)



Parallel system: the battery BMS cable goes to the BMS port of one inverter.

3.7 AC Input/Output Connection

CAUTION For safe operation and to meet code, install the equipment as follows.

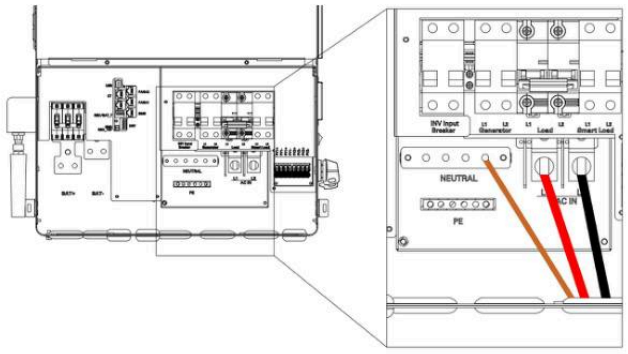
- Install a protective switching device between the utility grid and the inverter.
- The ratings of the wires and protective switching devices must meet or exceed the rated values of the equipment.
- Make sure all electrical connections are tight.

WARNING In parallel systems, you must get the polarity right during installation. Wrong polarity can cause a severe short circuit.

3.7.1 Recommended AC Input Cable and Breaker Specifications

Model	Wire size	Max bypass input current	Circuit breaker type	Torque value
KR-L12K-SP-BP	42.4 mm² (1 AWG) ×2P	200 A	3P—200 A	1.2-1.6 Nm (10.6-14.2 lbf·in)

AC Input Connection Diagram



AC input wiring (AC IN L1, L2 and NEUTRAL)

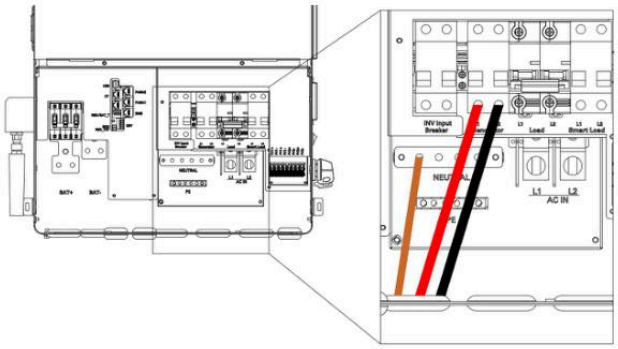
WARNING

- Turn OFF the power switch before you start any wiring.
- Always connect the protective earth (PE) wire first.
- Check the polarity markings closely so the circuit polarity is correct.

3.7.2 Recommended GEN Input Cable and Breaker Specifications

Model	Wire size	Maximum GEN input current	Circuit breaker type	Torque value
KR-L12K-SP-BP	13.3 mm² (6 AWG)	60 A	3P—63 A	1.2-1.6 Nm (10.6-14.2 lbf·in)

GEN Input Connection Diagram



Generator input wiring (Generator L1, L2 and NEUTRAL)

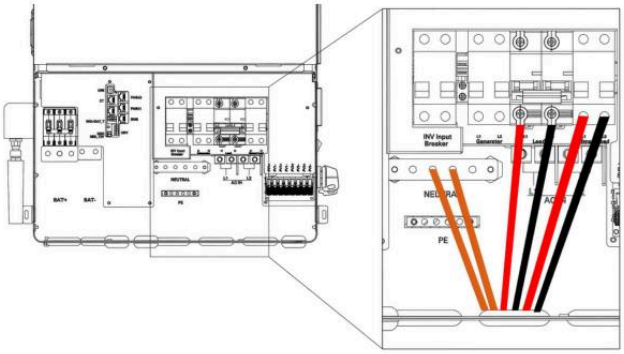
WARNING

- Turn OFF the power switch before you start any wiring.
- Always connect the protective earth (PE) wire first.
- Check the polarity markings closely so the circuit polarity is correct.

3.7.3 Recommended LOAD Cable and Breaker Specifications

Model	Wire size	Maximum load output current	Circuit breaker type	Torque value
KR-L12K-SP-BP	33.62 mm² (2 AWG)	125 A	3P—125 A	1.2-1.6 Nm (10.6-14.2 lbf·in)

LOAD Connection Diagram



Load output wiring

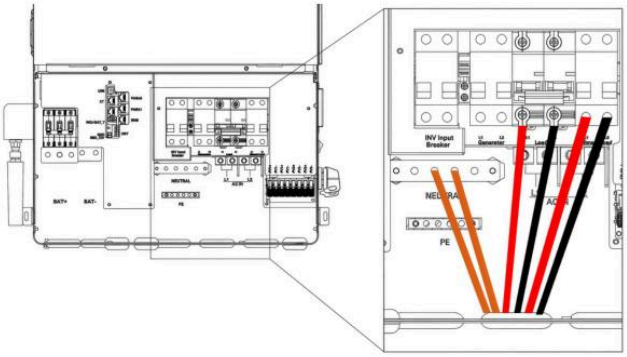
WARNING

- Turn OFF the power switch before you start any wiring.
- Always connect the protective earth (PE) wire first.
- Check the polarity markings closely so the circuit polarity is correct.

3.7.4 Recommended Smart Load Cable and Breaker Specifications

Model	Wire size	Maximum load output current	Circuit breaker type	Torque value
KR-L12K-SP-BP	13.3 mm² (6 AWG)	60 A	3P—63 A	1.2-1.6 Nm (10.6-14.2 lbf·in)

Smart Load Connection Diagram



Smart load output wiring

WARNING

- Turn OFF the power switch before you start any wiring.
- Always connect the protective earth (PE) wire first.
- Check the polarity markings closely so the circuit polarity is correct.

Notes for Special Loads

CAUTION An air conditioner must stay off for 2-3 minutes before it can safely restart. This lets the refrigerant pressure settle. Turning power back on right after an outage can damage the equipment. Before installation, check whether the air conditioner has a built-in start-up delay. If it does not, the high inrush current of an immediate restart may trip the off-grid solar inverter's overload protection again and again. This can damage internal parts of the air conditioner.

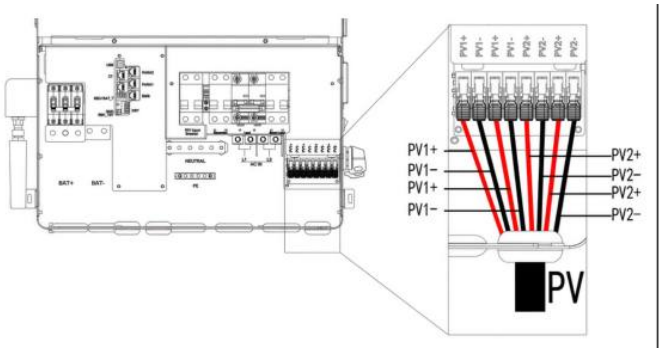
3.8 PV Connection

CAUTION For safe operation and to meet code, install the equipment as follows.

- Install a protective switch between the PV panels and the inverter.
- Wire and switch ratings must meet the equipment's specifications.
- Install a lightning protection device (LPD) on the inverter's DC input side.
- Make sure all electrical connections are tight.

Model	Wire size	Maximum PV input current	Circuit breaker type	Torque value
KR-L12K-SP-BP	5.2 mm² (10 AWG)	32 A	2P—40 A	1.2-1.6 Nm (10.6-14.2 lbf·in)

PV Connection Diagram



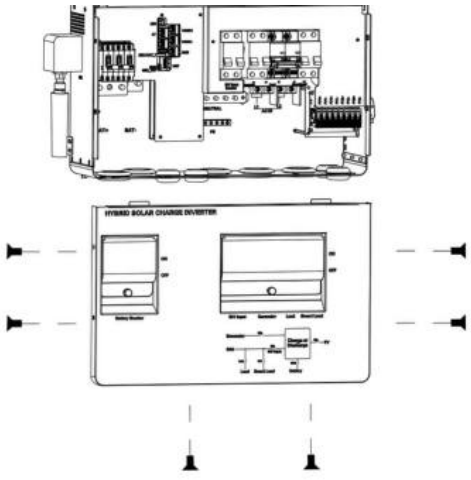
PV wiring: terminals PV1+, PV1-, PV1+, PV1-, PV2+, PV2-, PV2+, PV2-

WARNING

- Turn OFF the power switch before you start any wiring.
- Always connect the protective earth (PE) wire first.
- Check the polarity markings closely so the circuit polarity is correct.
- Do not connect one PV array to two inverters.
- Prevent system overvoltage so parts are not damaged.

3.9 Final Assembly

Finish all wiring connections. Then put the terminal cover back on and tighten the mounting screws securely.



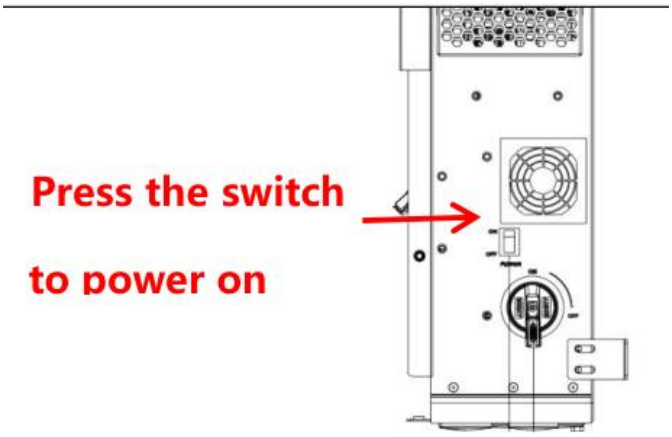
Reinstall the cover and its mounting screws.

4. Operation

4.1 Power ON/OFF

Press the switch to turn the inverter on.

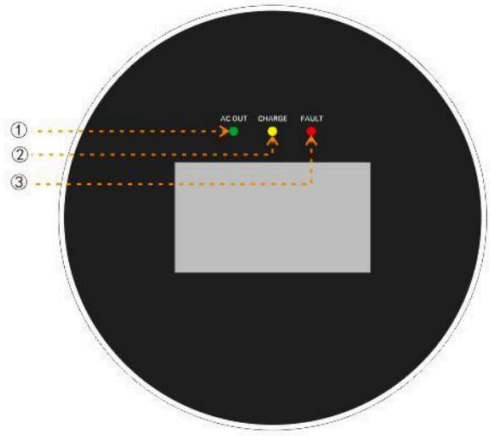
> **WARNING** Make sure all installation and wiring are correct before you turn the power on.



Press the switch to power on

4.2 Operation and Display Panel

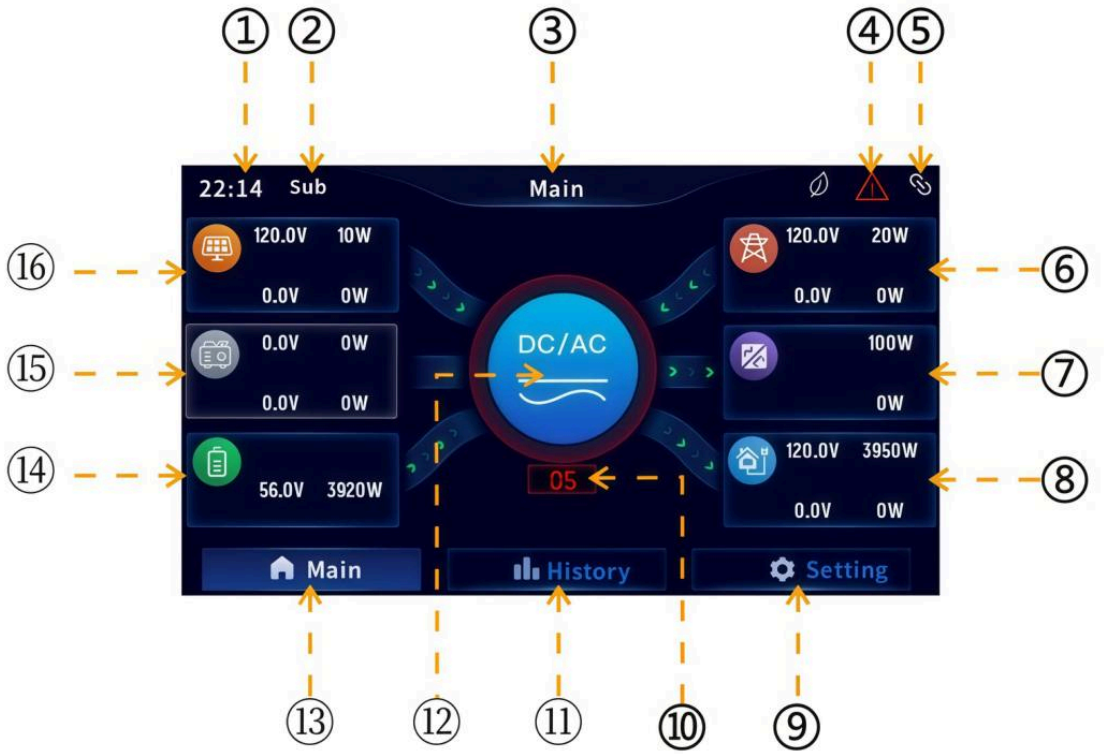
1) LED Indicator Lights



LED lights on the front panel: AC OUT, CHARGE, FAULT

No.	Light and meaning
①	**Output indicator (green):** Flashes when the inverter is supplying power.
②	**Charging indicator (yellow):** Flashes during charging.
③	**Fault indicator (red):** Lights up when a fault occurs. It either stays on steady or flashes. This shows how serious the fault is.

2) LCD Display and Parameter Configuration



LCD main screen with callout numbers

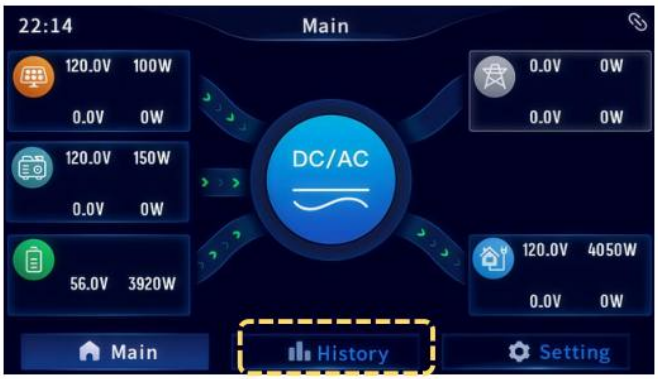
No.	Item	No.	Item
①	Time	⑨	Setting
②	Parallel Mode Tag	⑩	Alarm Code
③	Page Selection Note	⑪	Historical data
④	Alert Indicator	⑫	System information
⑤	LCD Communication Indicator	⑬	Main
⑥	Grid (AC input) parameters	⑭	Battery parameters
⑦	Normal Load	⑮	Generator parameters
⑧	Total Output Load	⑯	Photovoltaic (PV) parameters

3) Main Screen Icon Functions

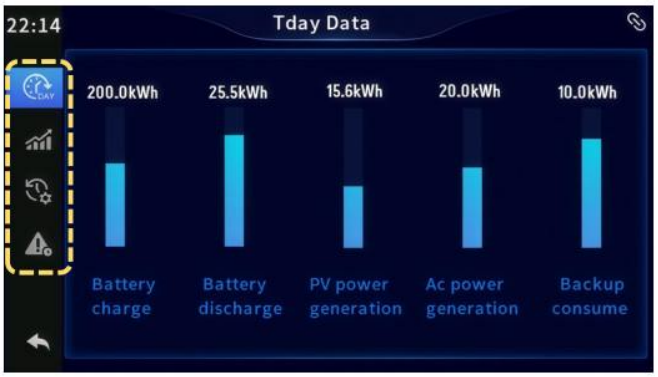
- The LCD main screen has icons for the main parts of the system, such as Battery, PV and Power Grid. Each icon shows its key values in real time. Tap an icon to open that part's details page and see all of its values.
- For example, tap the DC/AC icon in the center of the main screen to check the inverter's operating temperature, rated power, software version and other information.
- When the inverter has an alarm, the outer ring of the icon turns red, and an alarm code shows below it at the same time. Tap the alarm code to see the exact alarm type.
- The items below show every icon and what you can view with it.

A. History

Tap the History icon to view Today, Accumulation, Setting log and Fault log data.



Tap History



Today data page

B. Grid

Tap the grid icon to view L1/L2 values: V, A, VA, W, Hz.



Tap the grid icon



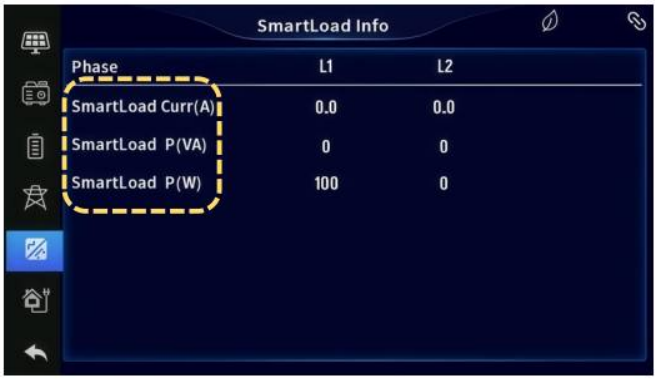
Grid Info page

C. Smart Load / Micro Inverter

Tap the Smart Load/MicInv Info icon to view A, VA and W.



Tap the Smart Load/MicInv icon



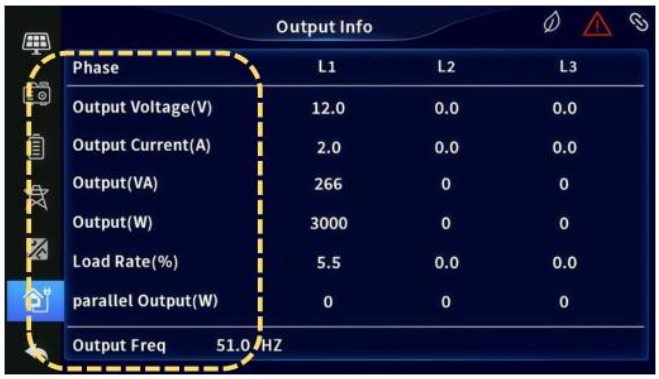
SmartLoad Info page

D. Total Output Load

Tap the Total Output Load icon to view L1/L2 values: V, A, VA, W, Load Rate, Hz.



Tap the Total Output Load icon



Output Info page

E. Alarm Code

Tap the Alarm Code icon or the Alert Indicator to view the device alarm log.



Tap the Alarm Code or Alert Indicator



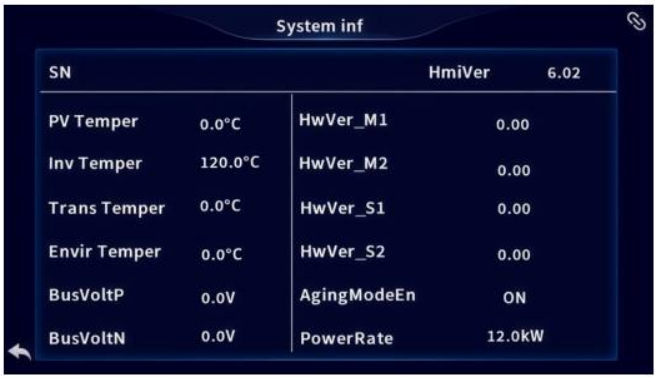
Current alarm page

F. System Information

Tap System information to view device details: Temp, Barcode, Power, Firmware, BUS V.



Tap the DC/AC (System information) icon



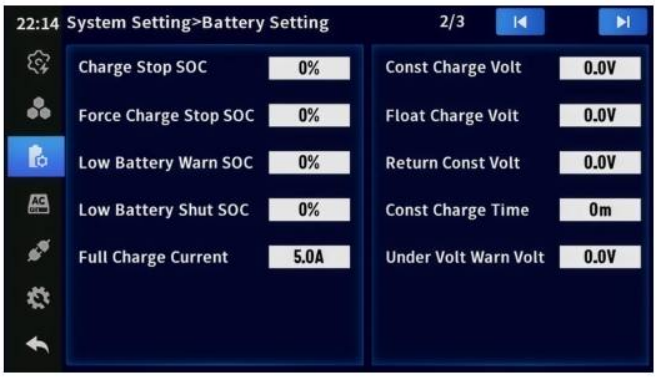
System inf page

G. Battery

Tap the Battery parameters icon to view Voltage, Charge/Discharge Current and SOC.



Tap the battery icon



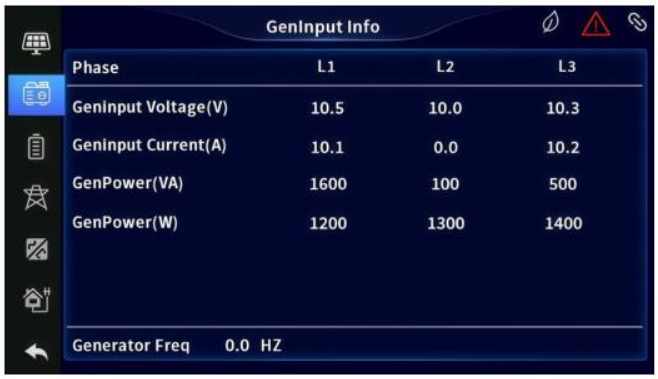
Battery page

H. Generator

Tap the Generator parameters icon to view L1/L2 values: V, A, Hz, VA, W.



Tap the generator icon



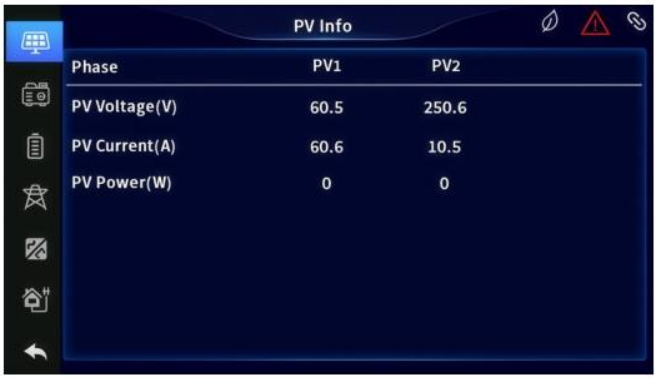
GenInput Info page

I. Photovoltaic (PV)

Tap the Photovoltaic (PV) parameters icon to view L1/L2 values: V, A, W.



Tap the PV icon



PV Info page

4) Setup Parameters Description

Step 1. Tap Setting on the main screen to open the settings page. You will see six main setting groups: Quick, EMS, Battery, Grid, Others and Advanced.



Tap Setting



Setting page with six groups

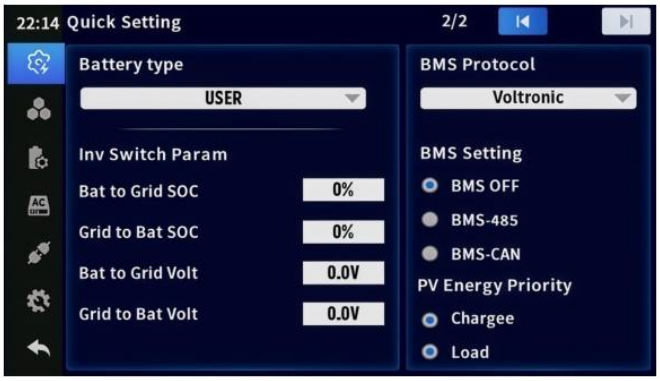
5) Settings Reference

All settings are sorted into groups and explained below. Read them carefully before you change any setting.

Quick



Quick Setting, page 1/2



Quick Setting, page 2/2

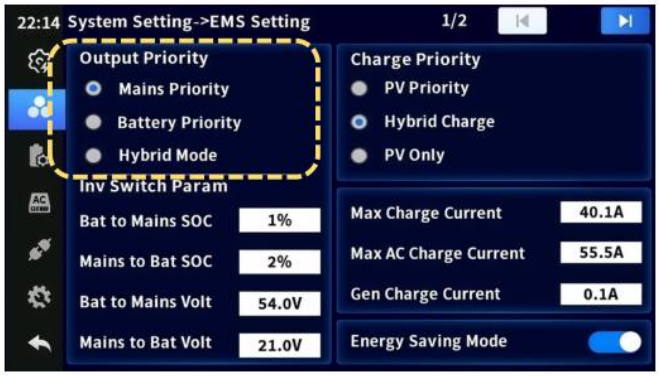
The Quick pages bring common settings together in one place. Each one is explained in its own group:

1. **Output Priority:** See the "EMS" settings.
2. **Energy Select:** See the "Advanced" settings.
3. **Timed Grid Charging** and **Timed Battery Discharge:** See the "EMS" settings.
4. **Battery Type / Inv Switch Param / BMS Protocol / BMS Setting:** See the "Battery" settings.
5. **PV Energy Priority:** See the "Advanced" settings.

EMS (Energy Management)

Output Priority

1. Mains Priority (default)
2. Battery Priority
3. Hybrid Mode



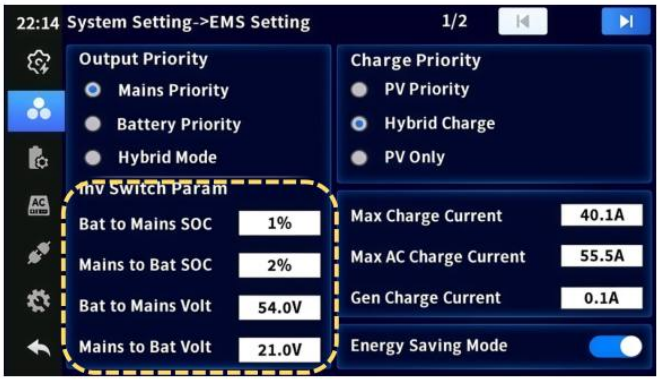
EMS Setting: Output Priority

1. **Mains Priority (default):** When the grid (mains) power is normal, the unit runs in grid (mains) mode. If grid power fails, it switches to battery inverter mode.
2. **Battery Priority:** When the grid is normal: if battery voltage drops below the "Bat to Mains Volt/SOC" setting, the unit switches to grid (mains) supply mode. If battery voltage rises above the "Mains to Bat Volt/SOC" setting, it switches to battery discharge mode. When the grid has a fault: the inverter stops output when battery voltage drops below "EOD Volt/Low Battery Shutoff".
3. **Hybrid Mode:** Select Hybrid Mode under Output Priority.

Inv Switch Param (points where the unit switches between battery and grid)

On the EMS screen these are shown as Bat to Mains SOC, Mains to Bat SOC, Bat to Mains Volt and Mains to Bat Volt.

1. Bat to Grid SOC: 10% default
2. Grid to Bat SOC: 100% default
3. Bat to Grid Volt: 46V default
4. Grid to Bat Volt: 57.6V default

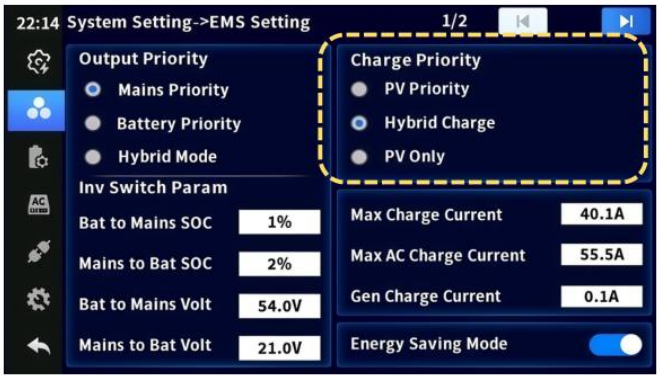


EMS Setting: Inv Switch Param

1. **Bat to Grid SOC (10% default):** When BMS is enabled, the system switches to grid (mains) power if the battery charge level (SOC) is below this value and grid power is available.
2. **Grid to Bat SOC (100% default):** This works in Battery Priority mode with BMS enabled. The system switches from grid (mains) mode to inverter mode when the battery charge level (SOC) goes above this value.
3. **Bat to Grid Volt (46V default):** In Battery Priority mode, the output switches from inverter to grid (mains) supply when battery voltage drops below this value. Setting range: 44V–52V.
4. **Grid to Bat Volt (57.6V default):** In Battery Priority mode (Output Priority set to Battery Priority), when battery voltage rises above this value, the output switches from the grid (mains) to the inverter. Setting range: 48V–60V.

Charge Priority

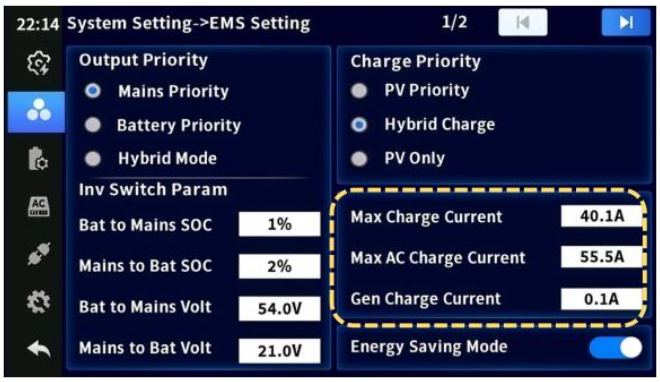
1. PV Priority
2. Hybrid Charge
3. PV Only



EMS Setting: Charge Priority

1. **PV Priority:** When there is enough PV (solar) power, the system uses it first to charge the battery, and extra power runs the loads. When PV power is not enough, battery charging still comes first, and the loads switch to grid supply.
2. **Hybrid Charge:** PV and grid charge together. PV charging comes first. When PV power is not enough, the grid adds charging power.
3. **PV Only:** PV-only charging. Only solar power charges the battery. The grid does not help.

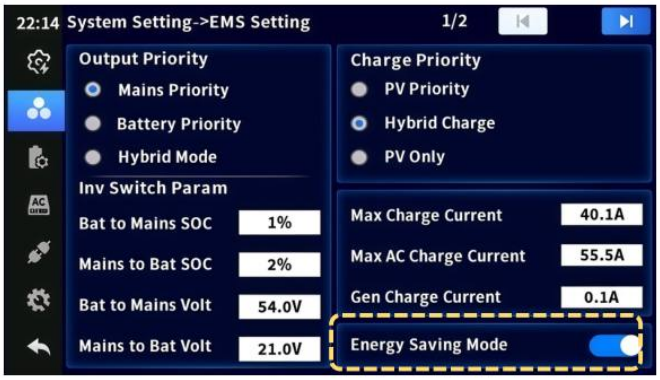
1. **Max Charge Current:** range 0–250A; 60A default
2. **Max AC Charge Current:** range 0–160A; 120A default
3. **Gen Charge Current:** range 0–250A; 60A default



EMS Setting: charge current limits

Power saving mode (Energy Saving Mode): Disabled by default.

When energy-saving mode is on and the load on each phase is under 25W, the inverter puts out a pulsed voltage after 5 minutes. If the load on any phase goes over 50W, normal output comes back on its own.



EMS Setting: Energy Saving Mode

Timed Grid Charging

- Disable (default) or Enable.

When enabled, the grid charges the battery and powers the load during the charging times you set, or when battery voltage is low.

- 1–3 Start Time (start charging, time slots 1 to 3): range 00:00–23:59
- 1–3 End Time (end charging, time slots 1 to 3): range 00:00–23:59



EMS Setting 2/2: Timed Grid Charging

Timed Battery Discharge

- Disable (default) or Enable.

When enabled, the battery discharges only during the discharge times you set, or when utility power is not available.

- 1–3 Start Time (time slots 1 to 3): range 00:00–23:59
- 1–3 End Time (time slots 1 to 3): range 00:00–23:59



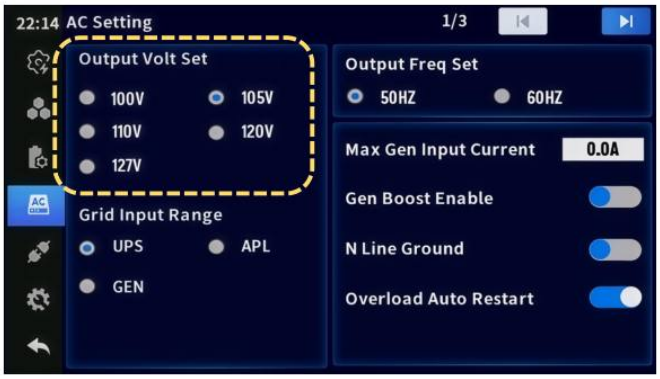
EMS Setting 2/2: Timed Battery Discharge

Grid (AC Setting)

Output Volt Set

- 120Vac default.

Can be set to 100Vac / 105Vac / 110Vac / 120Vac / 127Vac. Default is 120V. The rated output power is reduced as follows: (Power Rate) × (Vset / 120).



AC Setting: Output Volt Set

Output Frequency (Output Freq Set)

- 50.0Hz or 60.0Hz.

Grid-adaptive feature: When the grid is connected, the system matches the grid frequency on its own. When the grid is disconnected, you can set the output frequency in this menu. The default output frequency is 60Hz for 120Vac.



AC Setting: Output Freq Set

Grid Input Range

1. UPS (default)
2. Generator (GEN)
3. APL



AC Setting: Grid Input Range

1. **UPS (default):** Auto-detection when the grid is connected. Voltage range: 90–140Vac (when system voltage is 100V/105V, the minimum input voltage is limited to 85V). Frequency range: 47–55Hz (for 50Hz); 57–65Hz (for 60Hz).
2. **Generator:** Auto-detection is enabled when a diesel generator is connected. Voltage range: 90–140Vac. Frequency range: 40–70Hz.
3. **APL:** Turned on by the user. For both grid and generator: voltage range 85–140Vac; frequency range 40–70Hz.

1. **Max Gen Input Current:** 60A default. Range: 0–60A.
2. **Gen Boost Enable**
3. **N Line Ground**
4. **Overload Auto Restart**



AC Setting: generator and restart options

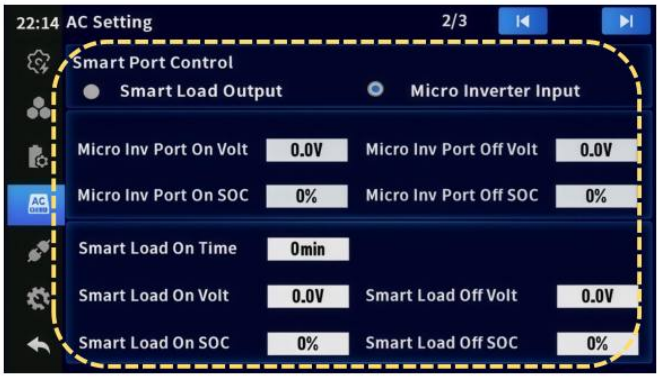
1. **Gen Boost Enable:** For what this does, see "Generator Breaker" (Solution 2) in Section 2.2.
2. **N Line Ground:** For how to use this, see "Neutral-to-Ground Connection" in Section 3.4 "System Wiring Diagram".

3. **Overload Auto Restart:** Turns on automatic restart after an overload. If the output shuts down from an overload, the unit restarts the output after a 3-minute delay. After a total of 5 restarts within 30 minutes, the system no longer restarts.

Smart Port Control

1. Smart Load Output
2. Micro Inverter Input

- For what this port does, see "Smart Load Breaker" in Section 2.2.



AC Setting 2/3: Smart Port Control

No.	Setting	Default	Setting range	Rule
1	**Micro Inv Port on Volt**	50V	40V–60V	Start voltage must be 1.2V lower than stop voltage before it can be set.
2	**Micro Inv Port Off Volt**	52V	40V–60V	Shutdown voltage must be 1.2V higher than start voltage before it can be set.
3	**Micro Inv Port on SOC**	20%	0%–100%	Start SOC must be at least 1% below the off (close) SOC before it can be set.
4	**Micro Inv Port Off SOC**	85%	0%–100%	Off (close) SOC must be at least 1% above start SOC before it can be set.
5	**Smart Load on Volt**	52V	40V–60V	Start voltage must be 1.2V above shutdown voltage before it can be set.
6	**Smart Load Off Volt**	50V	40V–60V	Shutdown voltage must be 1.2V below start voltage before it can be set.
7	**Smart Load on SOC**	85%	0%–100%	Start SOC must be at least 1% above the set stop SOC.
8	**Smart Load Off SOC**	20%	0%–100%	Shutdown SOC must be at least 1% below the set start SOC.

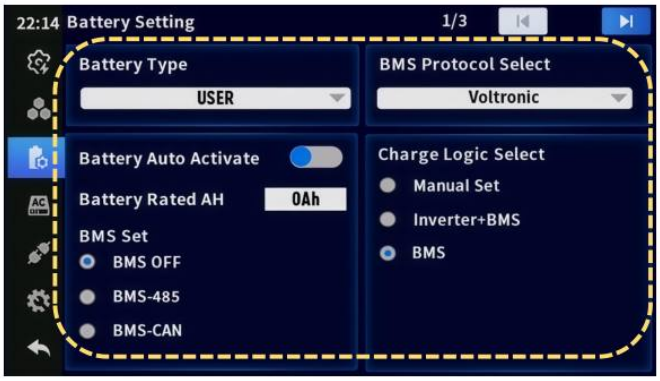
1. **Grid Zero Power Compensation** (screen: Grid 0 Power Comp): 0W default. Adjusts the grid power compensation.
2. **AC Couple Enable** (AC Coupling Enable): Disabled by default.
3. **Motor DriveOpt Enable:** Disabled by default. When enabled, helps inverter-type air conditioners start reliably.



AC Setting 3/3

Battery

1. **Battery Type**
2. **Automatic battery activation** (Battery Auto Activate)
3. **Battery rated capacity (Ah)** (Battery Rated AH)
4. **BMS Set** (BMS OFF default, BMS-485, BMS-CAN)
5. **BMS Protocol Select**
6. **Charge Logic Select** (Manual Set, Inverter+BMS, BMS)



Battery Setting 1/3

1. Battery Type

Type	Description
USER	User-defined. All battery settings can be changed.
SLD	Sealed lead-acid battery. Constant-voltage charge voltage: 57.6V. Float charge voltage: 55.2V.
FLD	Vented lead-acid battery. Constant-voltage charge voltage: 58.4V. Float charge voltage: 55.2V.
GEL (default)	Colloidal (gel) lead-acid battery. Constant-voltage charge voltage: 56.8V. Float charge voltage: 55.2V.
LFP15/LFP16	Lithium iron phosphate battery. LFP15 = 15 cells in series; LFP16 = 16 cells in series. For 16 cells in series, the default constant-voltage charge voltage is 56.8V. For 15 cells in series, the default constant-voltage charge voltage is 53.2V. Adjustable.
NCM14	Ternary lithium battery. Adjustable. The default constant-voltage charge voltage for the NCM 14 series is 57.6V.

2. Automatic battery activation

- When the battery is asleep (dormant) or not connected, PV or grid (mains) power wakes the battery up on its own and turns on the battery output.

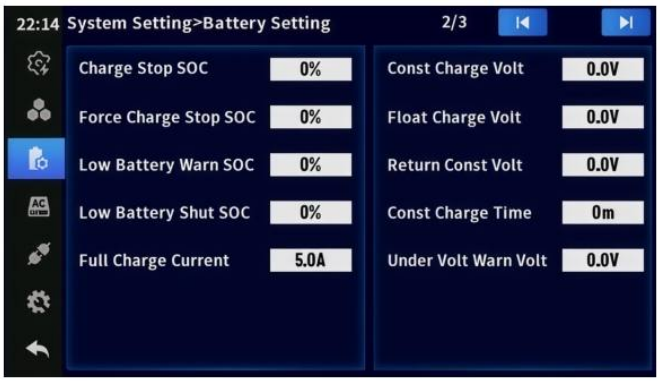
5. BMS Protocol Select

- When BMS communication is turned on, you must select the matching lithium battery maker's brand so they can communicate.

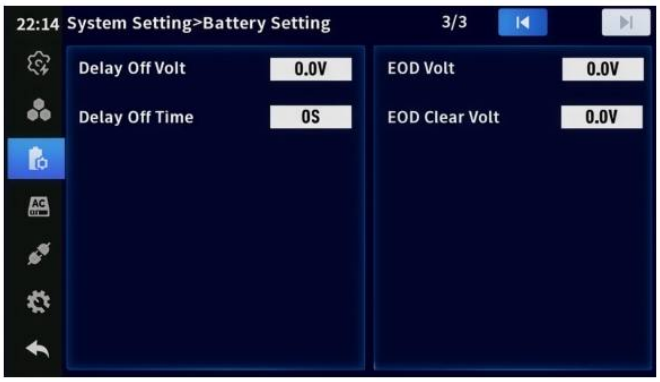
6. Charge Logic Select (Manual Set, Inverter+BMS, BMS)

- **Manual Set:** The battery's maximum charging current is limited to the value you set.
- **Inverter+BMS:** The battery's maximum charging current is limited by the inverter's built-in current-derating logic.
- **BMS:** The battery's maximum charging current is limited by the BMS current limit.

The screens below show more battery settings. Set them to match the battery you actually use and your needs.



Battery Setting 2/3: Charge Stop SOC, Force Charge Stop SOC, Low Battery Warn SOC, Low Battery Shut SOC, Full Charge Current, Const Charge Volt, Float Charge Volt, Return Const Volt, Const Charge Time, Under Volt Warn Volt



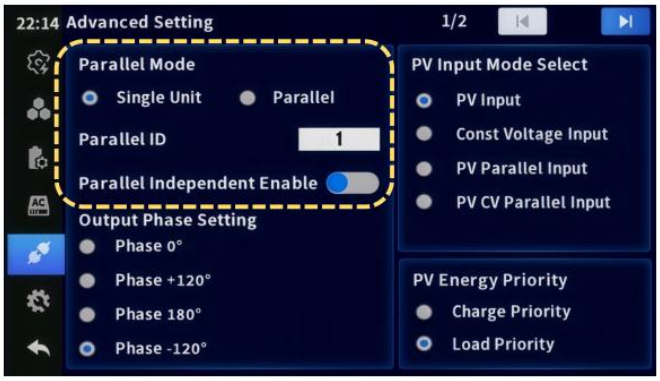
Battery Setting 3/3: Delay Off Volt, Delay Off Time, EOD Volt, EOD Clear Volt

Advanced

Parallel Mode

1. Single Unit
2. Parallel
3. Parallel ID
4. Parallel Independent Enable

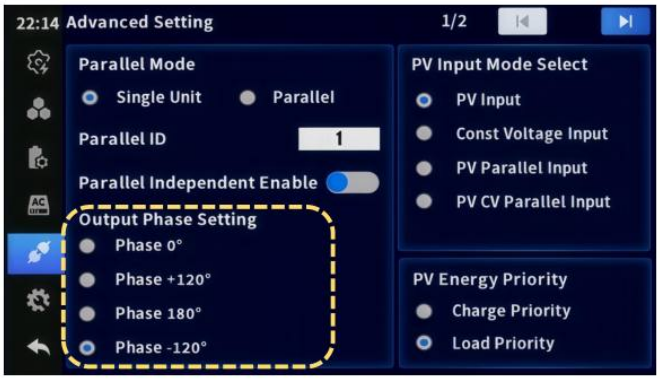
For parallel system wiring and setup, see Chapter 8: Parallel Installation, Wiring, and Setup.



Advanced Setting: Parallel Mode

Output Phase Setting

1. Phase 0°
2. Split 120° (screen: Phase +120°)
3. Split 180° (screen: Phase 180°)
4. Split -120° (screen: Phase -120°)



Advanced Setting: Output Phase Setting

- **Phase 0°:** L1 and L2 in parallel. Output is 120Vac L/N.
- **Split 120°:** L1 and L2 are 120° apart, using a tapped connection. L1-N / L2-N: 120Vac. L1-L2: 208Vac.
- **Split 180°:** L1 and L2 are 180° apart, with a tapped connection. L1-N / L2-N: 120Vac. L1-L2: 240Vac.
- **Split -120°:** L1 and L2 are -120° apart, with a tapped connection. L1-N / L2-N: 120Vac. L1-L2: 208Vac.

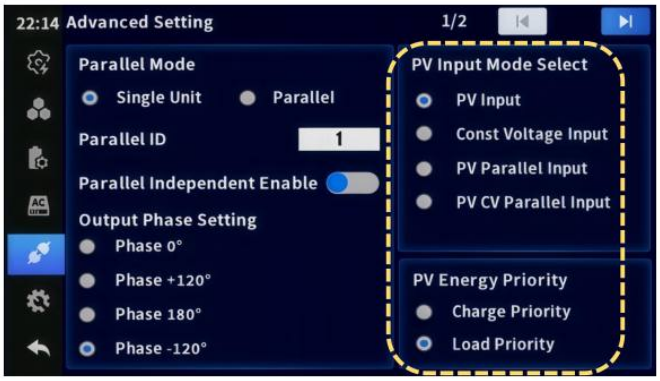
The user must set the output phase while the inverter is in standby mode. Make sure the inverter wiring is set up correctly. If it is not, the system may malfunction.

PV Input Mode Select

- PV Input
- Const Voltage Input
- PV Parallel Input
- PV CV Parallel Input

PV Energy Priority

- Charge Priority
- Load Priority

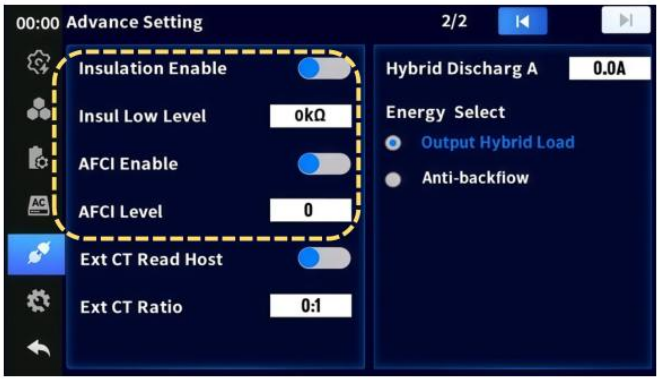


Advanced Setting: PV Input Mode Select and PV Energy Priority

PV Input Mode Select

1. **PV Input:** Two separate PV string systems, each connected to its own PV input.
2. **Const Voltage Input:** The two PV inputs are each connected to their own DC constant-voltage power supply.
3. **PV Parallel Input:** The two PV inputs can be connected in parallel to one PV string system.
4. **PV CV Parallel Input:** The two PV inputs are connected in parallel to one DC constant-voltage source system.

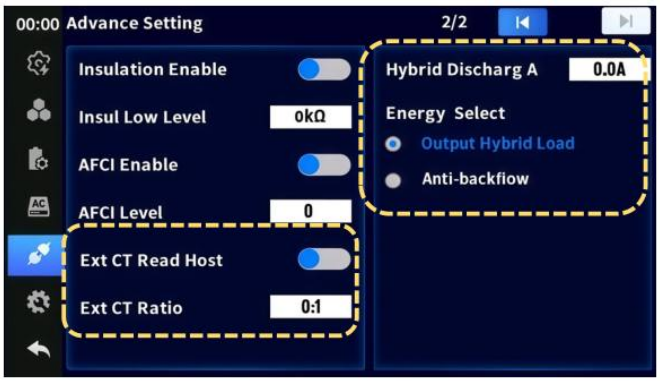
1. **Insulation Enable**
2. **Insul Low Level:** 15kΩ. Setting range: 0–3MkΩ.
3. **AFCI Enable:** Disabled by default.
4. **AFCI Level**



Advance Setting 2/2: insulation and AFCI

1. **Insulation Enable:** MPPT runs only if the inverter detects a PV grounding impedance below the set threshold.
2. **AFCI Enable** (arc-fault protection): When turned on, if the inverter detects a PV arc fault, it cuts off the RSD_12V power supply and shuts down the MPPT system.

1. **Hybrid Discharge A** (Hybrid Discharg A): 272A default. Setting range: 0–272A.
2. **External CT Get Host Enable** (Ext CT Read Host)
3. **External CT Ratio** (Ext CT Ratio): 2000:1 default. Can be set to match your actual needs.
4. **Energy Select**
   - Output Hybrid Load
   - Anti-backflow



Advance Setting 2/2: Hybrid Discharg A, Energy Select, Ext CT

External CT Get Host Enable

- **Disable (default):** Controls only the grid-connected power of this inverter.
- **Enable:** This inverter is the anti-backflow host. Plug the CT into this inverter to control the total grid-connection power.

Energy Select

- **Output Hybrid Load:** The inverter sends the energy it produces only to the load (generation power < load power). Note: in mixed-load mode, grid (mains) and PV power the load together; when PV is not enough, the grid (mains) makes up the rest.
- **Anti-backflow:** Prevents reverse current. The system uses its power for its own loads, based on load power, and does not send current into the grid (mains).

Others

1. **Power Control:** Controls the inverter AC output.
2. **Device Reboot:** Restarts the inverter.
3. **System Time Set**
4. **Language Set:** Chinese or English.
5. **Factory Reset:** Factory Reset, Clear Stats Data, Clear History Data.
6. **Buzzer Mute**
7. **Screen Bright** (Screen Brightness)
8. **Screen Bright Times:** 5/10/15/20 Min



Other Setting

5. Operating Modes

5.1 Charging Mode

The charging mode sets where the battery gets its charge. There are three choices.

1) PV Priority

The solar (PV) modules charge the battery first. The grid/utility (mains) charges the battery only when the PV system fails. In the daytime, solar power is used fully for charging. At night, charging switches to the grid (mains). This mode keeps the battery charged. It works best where the grid is fairly stable and electricity is fairly expensive.



Solar Sufficient



Solar Insufficient

2) Hybrid Charge

PV and the grid (mains) charge the battery together. PV MPPT charging comes first. When PV power is not enough, the grid (mains) makes up the difference. When PV power is enough again, the grid (mains) stops charging. This is the fastest charging mode. It suits areas where the power grid is unstable, so you have enough backup power at any time.



Solar Sufficient



Solar Insufficient

3) PV Only

Only PV charges the battery. The grid (mains) does not charge it. This is the most energy-efficient mode, because only the solar panels charge the battery. It is usually used in areas with good sunlight.



PV Only

5.2 AC Output Mode

The AC output mode sets which power source feeds your loads first.

PV Priority Mode

Solar (PV) and the batteries power the load. The unit switches to the grid (mains) when PV charging fails. This mode uses as much solar energy as possible while keeping the battery charged. It suits areas where the grid is fairly stable.

Power supply priority: Solar → Battery → Utility (Solar Sufficient)
Power supply priority: Solar → Utility → Battery (Solar Insufficient)



Solar Sufficient



Solar Insufficient

Mains Priority Mode

The unit switches to the inverter only when the grid (mains) fails. When grid (mains) power is present, the unit switches to the grid for charging and for powering the loads. In this mode the unit works like a backup UPS. It suits areas where the grid is unstable. Switching does not affect PV charging.

Power supply priority: Utility → Solar → Battery



Utility Valid



Utility Outage

Battery Priority Mode

The unit switches to the grid (mains) only when the battery discharges below its low-voltage (undervoltage) set point ("Bat to Mains Volt/SOC" in Inv Switch Param, Section 4.2). When the charging battery rises above the "Mains to Bat Volt/SOC" set point, the unit switches back to battery discharge mode. This lets the battery cycle between charge and discharge. This mode uses as much DC power as possible. It is used in areas with a stable grid. Switching does not affect PV charging.

Power supply priority: Solar → Battery → Utility



Battery Available



Battery Under-voltage

Hybrid Mode

Constant Power On Grid: When PV generation is more than enough for the load, the extra power is fed to the grid as grid-connected generation.

Energy To Output: The energy the inverter generates goes only to the load (generation power < load power).

> **NOTE** Mixed load: the grid (mains) and PV carry the load together. When PV is not enough, the grid (mains) makes up the difference.

Curr Anti Rev (CT): Anti-backfeed (anti-reverse current). The unit powers your own loads based on the load power and does not send current into the grid (mains).

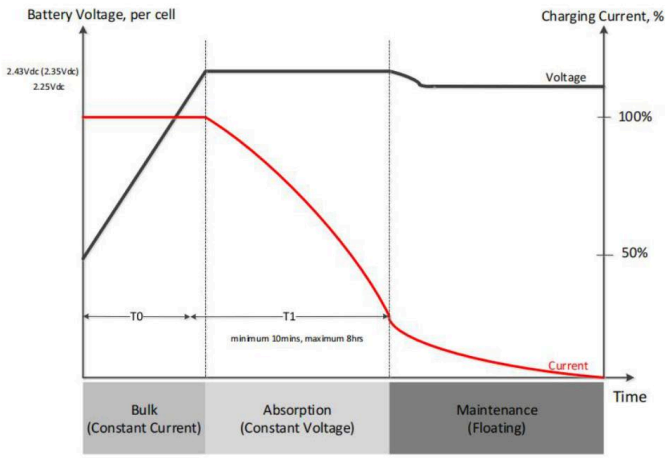


Curr Anti Rev (CT)

6. Battery Type Parameters

6.1 Lead-Acid Battery

Parameter \ Battery type	Sealed lead-acid battery (SLD)	Colloidal (gel) lead-acid battery (GEL)	Vented lead-acid battery (FLD)	User-defined (User)
Overvoltage disconnection voltage	60V	60V	60V	40~60V (Adjustable)
Battery fully charged recovery point	52V (Adjustable)	52V (Adjustable)	52V (Adjustable)	52V (Adjustable)
Boost charge voltage	57.6V	56.8V	58.4V	40~60V (Adjustable)
Floating charge voltage	55.2V	55.2V	55.2V	40~60V (Adjustable)
Undervoltage alarm voltage (01 fault)	44V	44V	44V	40~60V (Adjustable)
Undervoltage alarm voltage recovery point (01 fault)	Undervoltage alarm voltage + 0.8V			
Low voltage disconnection voltage (02 fault)	42V	42V	42V	40~60V (Adjustable)
Low voltage disconnection voltage recovery point (02 fault)	52V (Adjustable)	52V (Adjustable)	52V (Adjustable)	52V (Adjustable)
Discharge limit voltage	40V	40V	40V	40~60V (Adjustable)
Over-discharge delay time	5s	5s	5s	1~30s (Adjustable)
Boost charge duration	120 minutes	120 minutes	120 minutes	10~600 minutes (Adjustable)
Lead-acid charging curve	See the chart below.			



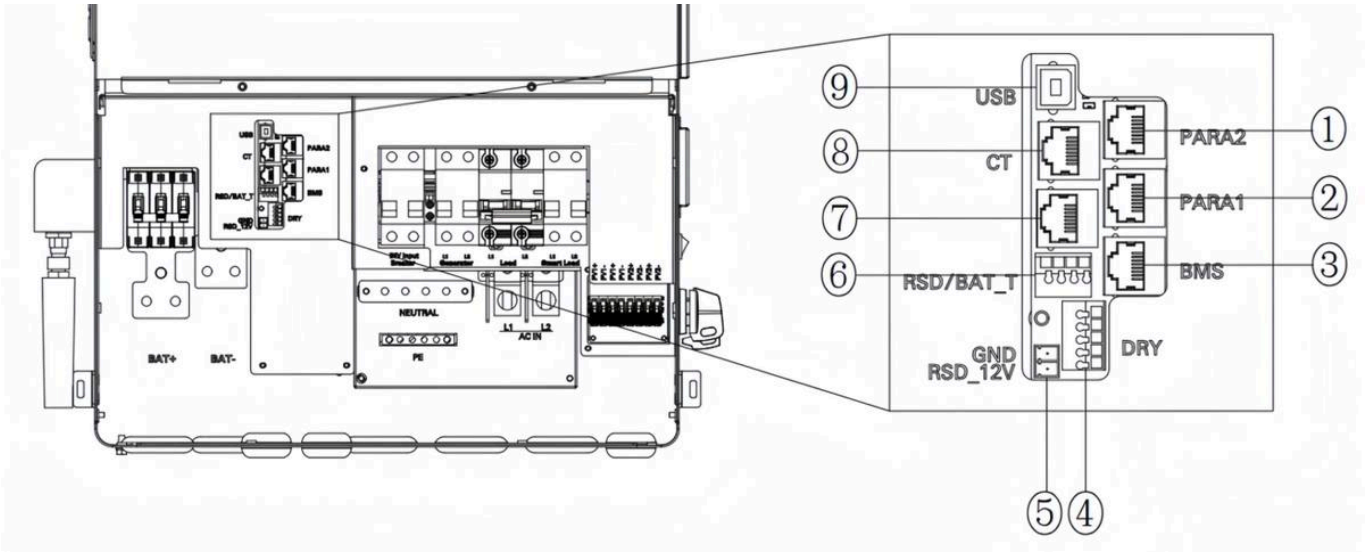
Lead-acid charging curve: Bulk (Constant Current, T0), Absorption (Constant Voltage, T1: minimum 10 mins, maximum 8 hrs), Maintenance (Floating)

6.2 Lithium Battery

Parameter \ Battery type	Ternary lithium battery (N14)	Lithium iron phosphate battery (LF16)	Lithium iron phosphate battery (LF15)	Lithium iron phosphate battery (LF14)
Overvoltage disconnection voltage	60V	60V	60V	60V
Battery fully charged recovery point	54.8V (Adjustable)	53.6V (Adjustable)	50.4V (Adjustable)	47.6V (Adjustable)
Boost charge voltage	57.6V (Adjustable)	56.8V (Adjustable)	53.2V (Adjustable)	49.2V (Adjustable)
Floating charge voltage	57.6V (Adjustable)	56.8V (Adjustable)	53.2V (Adjustable)	49.2V (Adjustable)
Undervoltage alarm voltage (01 fault)	46.8V (Adjustable)	49.6V (Adjustable)	46.4V (Adjustable)	43.2V (Adjustable)
Undervoltage alarm voltage recovery point (01 fault)	Undervoltage alarm voltage + 0.8V			
Low voltage disconnection voltage (04 fault)	42V (Adjustable)	48.8V (Adjustable)	45.6V (Adjustable)	42V (Adjustable)
Low voltage disconnection voltage recovery point (04 fault)	49.6V (Adjustable)	52.8V (Adjustable)	49.6V (Adjustable)	46V (Adjustable)
Discharge limit voltage	39.2V	46.4V	43.6V	40.8V
Over-discharge delay time	30s (Adjustable)	30s (Adjustable)	30s (Adjustable)	30s (Adjustable)
Boost charge duration	120 minutes (Adjustable)	120 minutes (Adjustable)	120 minutes (Adjustable)	120 minutes (Adjustable)

7. Communication

7.1 Overview



Communication ports

No.	Port	No.	Port
①	Parallel_2	⑥	RSD Input Control/BAT_T
②	Parallel_1	⑦	
③	BMS Port	⑧	CT Port
④	DRY	⑨	USB Port
⑤	RSD_12V Power		

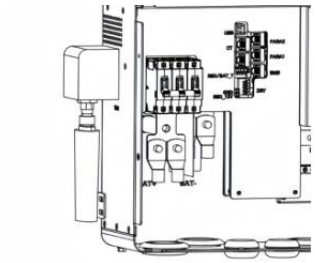
7.2 USB Port



USB Type B

- Used for USB communication with the optional PC host software.
- The computer must have the "USB to Serial Port Chip CH340T Driver" installed.

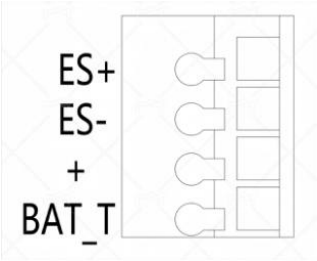
7.3 Wi-Fi Port



Wi-Fi port location

- The Wi-Fi/GPRS module converts the 485 signal.
- You can view operating data and change parameters remotely from a mobile phone or computer.
- Optional. It can be installed through the RJ45 interface (the RJ45 and USB ports are compatible; choose one) or through the aviation plug interface.

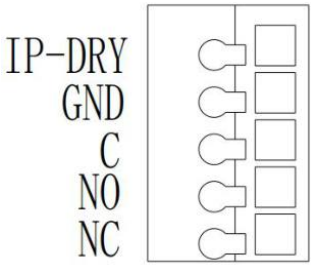
7.4 Rapid Shutdown System



RSD terminal: ES+, ES-, +, BAT_T

- The 12V DC output at the RSD+ and RSD- terminals shuts off (is de-energized) when either of these happens:
  - The ES+ and ES- terminals are connected to an emergency push-button switch, and the switch is closed; or
  - 12V DC is applied to the + and - terminals.

7.5 Dry Contact Port



Dry contact terminal: IP-DRY, GND, C, NO, NC

N-NO-NC function: Controls diesel generator start/stop

1. Starting state: NC-N is closed and NO-N is open.
2. When the battery voltage falls below the grid-switching threshold, NC-N opens and NO-N closes. The NO-N contact can drive resistive loads rated 125VAC/1A, 230VAC/1A, or 30VDC/1A.

IP-GND function: Remote control of the inverter AC output

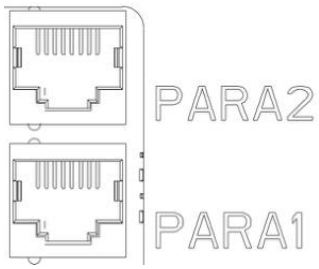
1. IP-GND disconnected: AC output ON.
2. IP-GND connected: AC output OFF.

7.6 CT Sampling Detection (Optional Feature)

The CT watches the inverter output current in real time to stop reverse power flow into the grid.

WARNING This inverter has not obtained the required grid-connection certification. It is limited to off-grid use only. Any unauthorized connection to the public grid violates local laws and regulations. For specific compliance requirements, consult a qualified lawyer in your area. KOOR expressly disclaims all liability for any legal consequences or damages caused by unauthorized grid-tied operation.

7.7 Parallel Communication Function



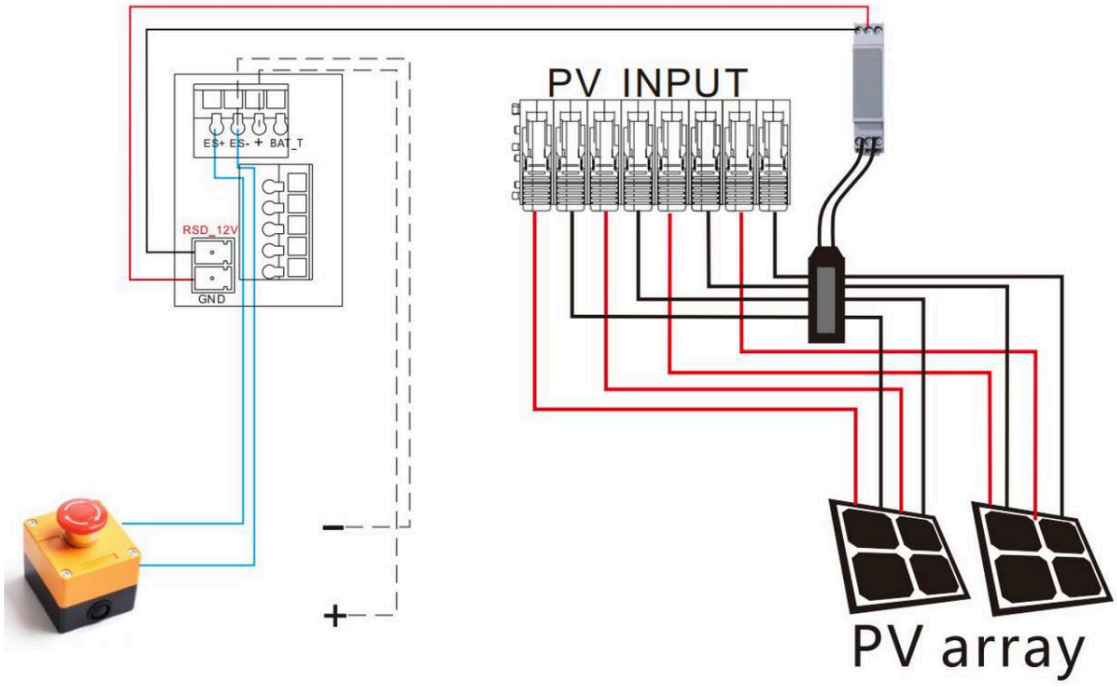
PARA2 and PARA1 ports

- The unit has two RJ45 ports: PARA_1 and PARA_2.

7.8 RSD System Introduction

The 12V supply from the RSD to the TRANSMITTER-PLC is cut off when either of these happens:

- The ES button (wired between ES+ and ES-) is activated; OR
- A 12V signal is applied to the + and - terminals.



RSD system wiring: emergency stop button, RSD terminal block, PV INPUT and PV array

8. Parallel Installation, Wiring, and Setup

CAUTION For safe operation and to meet code requirements, install the equipment as described below.

- You can connect up to **six (6)** inverters in parallel.
- Check your connections against the wiring diagrams below, both before and after you connect the system.
- If you add or remove an inverter in a parallel system that is running, turn off **all** power switches before you do any wiring.
- Check every connection before you power on. Wrong wiring can make the parallel system work abnormally.

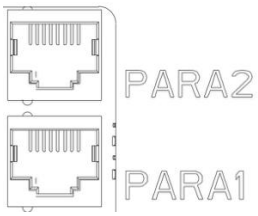
8.1 Power Wiring Precautions for Parallel Systems

Connection	What to do
Battery wiring	• Connect all inverters to the **same** battery bank. • Connect BAT+ to BAT+ and BAT- to BAT-. The positive and negative wires must be the same length and gauge. • Use the same wire length and gauge for every inverter.
GRID IN wiring	**Parallel connection in split phase:** • Connect L1 to L1, L2 to L2, N to N, and PE to PE on all inverters. **Parallel connection in single phase:** • Make sure every inverter is connected correctly, with no faults. Connect the L1 and L2 outputs in parallel to their own buses, N to N, and PE to PE. • Use the same wire length and gauge for all cables that run between the inverters.
GEN IN wiring	**Parallel connection in split phase:** • Connect L1 to L1, L2 to L2, N to N, and PE to PE on all inverters. **Parallel connection in single phase:** • Make sure every inverter is connected correctly, with no faults. Connect the L1 and L2 outputs in parallel to their own buses, N to N, and PE to PE. • Use the same wire length and gauge for all cables that run between the inverters.
AC OUT LOAD / SMART LOAD wiring	**Parallel connection in split phase:** • Connect L1 to L1, L2 to L2, N to N, and PE to PE on all inverters. **Parallel connection in single phase:** • Make sure every inverter is connected correctly, with no faults. Connect the L1 and L2 outputs in parallel to their own buses, N to N, and PE to PE. Use the same wire length and gauge for all cables that run between the inverters.
PV IN wiring	• On every inverter, each PV input must connect to its own independent, separate PV (solar) array.

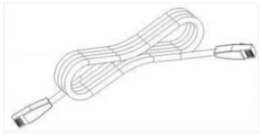
8.2 Parallel Communication Cable Precautions

1) Wiring the Parallel Communication Cable

- Each inverter has two RJ45 ports: **PARA_1** and **PARA_2**.
- Connect inverters to each other **only** from a PARA_1 port to a PARA_2 port. Never connect two ports of the same type (for example, PARA_1 to PARA_1, or PARA_2 to PARA_2) on any inverters.
- All connections must be secure and reliable.



PARA_2 and PARA_1 communication ports

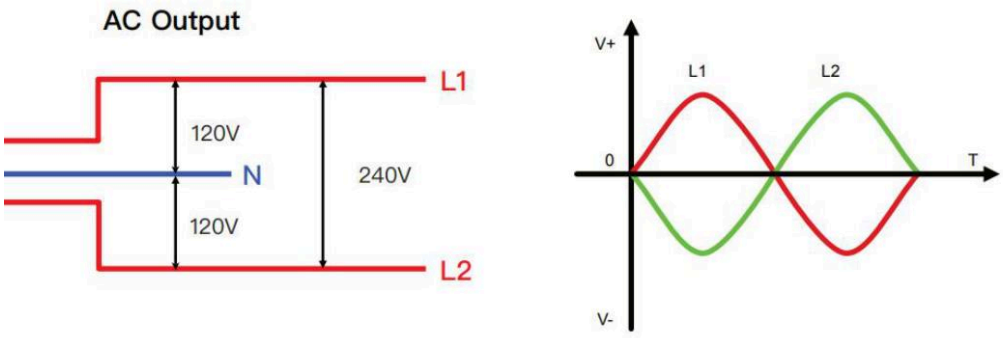


Parallel communication cable

8.3 Parallel Connection Schematic Diagrams

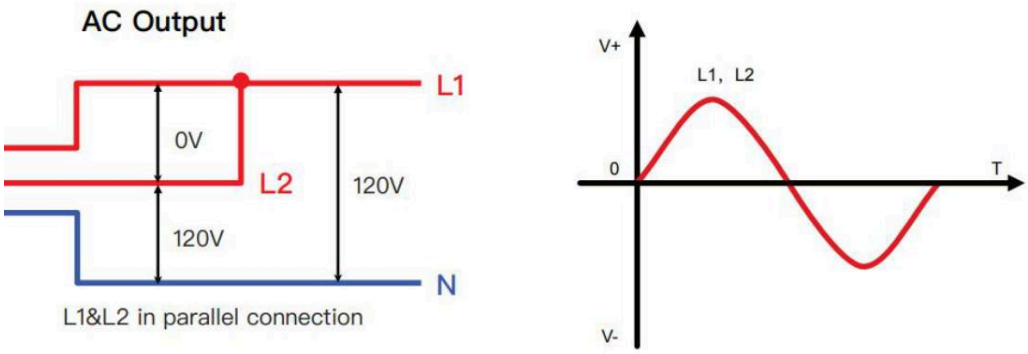
When several inverters run in parallel, connect them as shown in the diagrams below.

Parallel Connection in Split Phase



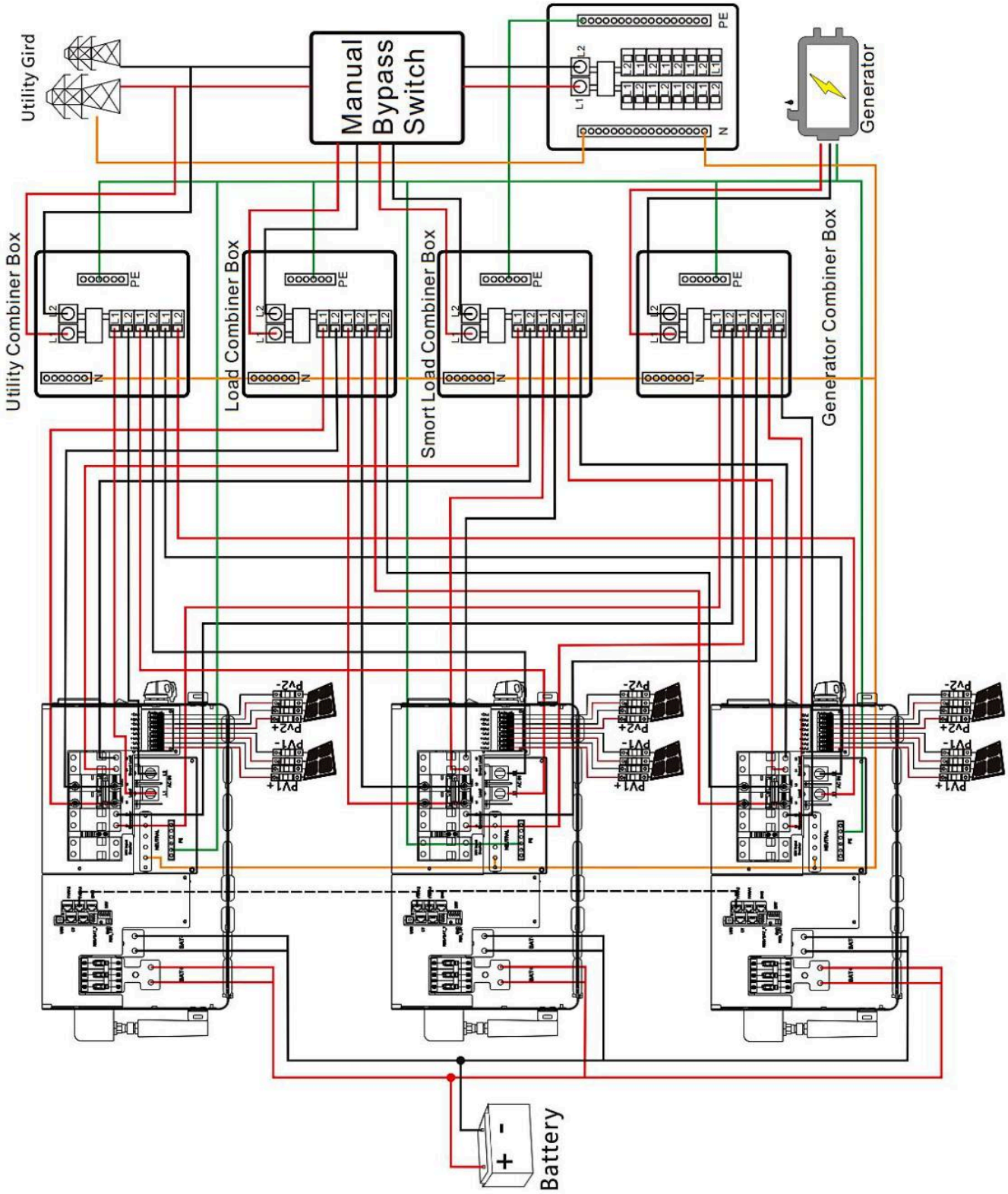
Split phase AC output: 120V from L1 to N, 120V from N to L2, and 240V from L1 to L2. L1 and L2 waveforms are opposite each other.

Parallel Connection in Single Phase



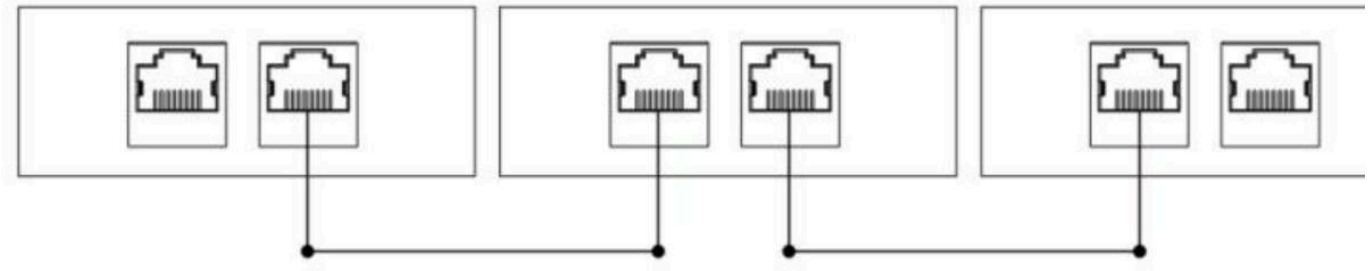
Single phase AC output: L1 and L2 connected in parallel. 0V between L1 and L2, and 120V from L1/L2 to N. L1 and L2 share one waveform.

Three Inverters Connected in Split-Phase Parallel (Up to 6 Inverters Can Be Paralleled)



Three-inverter split phase parallel system: utility grid, generator, manual bypass switch, breaker panel, utility / load / smart load / generator combiner boxes, PV1 and PV2 arrays on each inverter, and one shared battery bank

Parallel Communication Cable Connection Diagram



Communication cables run from one inverter to the next, port to port, in a chain

8.4 Parallel System Setup

CAUTION When you start the inverters to set up a parallel system, follow the steps below and these precautions.

- Make sure every inverter in the parallel system has its own unique AC/generator input.
- Make sure the output voltage and frequency settings are the same on all parallel inverters.
- To set the phase sequence, set up each inverter one at a time, with all the other inverters powered off. After every inverter has been set, you can restart the whole system.

CAUTION If you change the phase sequence, you must rewire the system to match the new setup.

- When the system is running, measure the output voltage and confirm it is correct. Then connect the loads.

8.4.1 Parallel System Setup Procedure

Step 1: Output Voltage/Frequency Settings



*A. Tap **Setting**.*



*B. Tap **Grid**.*

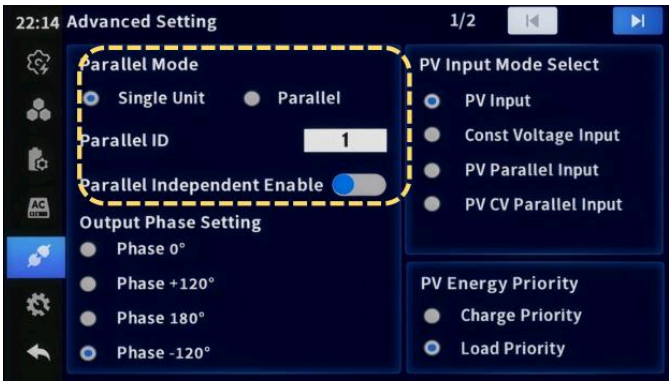


*C. Tap the AC output phase voltage/frequency settings (**Output Volt Set** and **Output Freq Set**).*

Step 2: Parallel Mode Setup



*A. Tap **Advanced**.*



*B. Tap **Parallel Mode** and **Parallel ID**.*

C. Select **Parallel**, then confirm.

D. Tap **Communication address**.

9. Fault Codes and Troubleshooting

9.1 Fault Codes

When the inverter finds a problem, it shows a fault code on the screen. The table below lists each code. The "Turns off output?" column tells you if the inverter shuts off its output for that fault.

Fault code	Fault name	Turns off output?	Description
[01]	Battery under voltage warning	No	The battery voltage is lower than the "battery under voltage alarm" setting. The battery is in an undervoltage state.
[02]	Battery under voltage protection	Yes	The battery voltage is low. The inverter turns off the output and stops the battery from discharging, to protect it.
[03]	Average battery discharge overcurrent protection	Yes	The average battery discharge current was higher than the maximum battery input current for 1 minute. The inverter turns off the output and stops the battery from discharging, to protect it.
[04]	Instantaneous battery discharge overcurrent protection	Yes	The instantaneous (moment-to-moment) battery discharge current is higher than the device's maximum instantaneous value. The inverter turns off the output and stops the battery from discharging, to protect it.
[05]	Battery not connected	Yes	Warning: the battery is not connected.
[06]	Battery over voltage	Yes	The battery voltage is higher than the limit for the selected battery type or the set battery voltage. The inverter turns off the output and stops charging the battery, to protect it.
[07]	BMS low battery alarm	No	The BMS (battery management system) reports a low battery alarm. (BMS must be enabled for this to work.)
[08]	BMS low battery protection	Yes	The BMS reports that the battery capacity (charge level) is low. The inverter turns off the output and stops the battery from discharging, to protect it. (BMS must be enabled for this to work.)
[09]	Bypass overload protection	Yes	The grid/utility (mains) side is overloaded. The inverter turns off the AC output and stops charging from the grid.
[10]	Battery output overload protection	Yes	The inverter is overloaded while running from the battery. The inverter turns off the AC output and stops the battery from discharging, to protect it.
[11]	Battery inverter output short circuit	Yes	The AC output is short-circuited while the inverter runs from the battery. The inverter turns off the AC output and stops the battery from discharging, to protect it.
[12]	The AC output of the battery inverter over circuit	Yes	The AC output is "over circuit" while the inverter runs from the battery. The inverter turns off the AC output and stops the battery from discharging, to protect it.
[13]	The DC component of the battery inverter voltage is abnormal	Yes	The DC component of the inverter output voltage (while running from the battery) is abnormal. The inverter turns off the AC output and stops the battery from discharging, to protect it.

Fault code	Fault name	Turns off output?	Description
[14]	Bus over voltage software sampling protection	Yes	The internal battery boost bus voltage is too high (software protection). The inverter turns off the AC output and charging.
[15]	Bus over voltage hardware sampling protection	Yes	The internal battery boost bus voltage is too high (hardware protection). The inverter turns off the AC output and charging.
[16]	Bus under voltage protection	Yes	The internal battery boost bus voltage is too low (undervoltage protection). The inverter turns off the AC output and charging.
[17]	Bus short circuit protection	Yes	The internal battery boost bus has a short circuit (short-circuit protection). The inverter turns off the AC output and charging.
[18]	The PV input voltage is over voltage	Yes	Protection: the solar input voltage is higher than the maximum allowed input voltage.
[19]	PV Over Curr Sw	Yes	Solar charging overcurrent (software protection). Solar charging is turned off.
[20]	PV Over Curr Hw	No	Solar charging overcurrent (hardware protection). Solar charging is turned off.
[21]	PV ISO Low	Yes	The PV insulation resistance (impedance) is low. The inverter shuts down.
[22]	The PV heat sink is overheated	No	The solar charging heat sink (radiator) is too hot. Solar charging is turned off.
[23]	The AC heat sink is overheated	Yes	The heat sink is too hot. AC charging or battery inverter discharge is turned off.
[24]	The temperature of the main transformer is overheated	Yes	The internal main transformer is too hot. AC charging or battery inverter discharge is turned off.
[25]	AC input relay short circuit	Yes	AC input relay short-circuit protection. It keeps the inverter's AC output from being pushed back into the bypass AC input. The inverter is turned off.
[26]	Output relay short	Yes	Output relay short circuit.
[27]	Fan failure	Yes	The fan is blocked or has failed. The inverter output and charging functions are turned off.
[28]	EEPROM	Yes	The program is not activated.
[29]	SPI Comm Err	Yes	Communication error between the master chip and the slave chip.
[30]	Type detection error	Yes	The model was not set before delivery, and the model identification is wrong.
[31]	Bus Soft Start Fail	Yes	The inverter busbar soft start failed.
[32]	LeakageCurr Over	Yes	The leakage current sensor has a sampling error.
[33]	Parallel control CAN communication is faulty	Yes	In parallel mode, CAN communication is lost. AC output and charging are turned off.
[34]	Parallel control CAN communication is faulty	Yes	In parallel mode, CAN communication is lost. AC output and charging are turned off.

Fault code	Fault name	Turns off output?	Description
[35]	Parallel mode is faulty	Yes	In parallel mode, the machines in the system have different Parallel Mode settings.
[36]	Parallel current sharing fault	Yes	In parallel mode, the battery inverter AC output current of each machine is very uneven (the machines do not share current evenly). The AC output and charging are turned off.
[37]	Parallel ID setting error	Yes	In parallel mode, the RS485 addresses conflict (repeat). This fault stops the AC output and charging. After the host automatically reassigns the addresses, the fault clears and the host enters parallel operation.
[38]	Inconsistent battery in parallel mode	Yes	In parallel mode, the battery input voltage is very different from machine to machine.
[39]	Inconsistent AC input source in parallel mode	Yes	The AC input source is not the same across machines in parallel mode.
[40]	The parallel mode synchronization fails	Yes	In parallel mode, the hardware synchronization signal was not received. Parallel operation and AC output stop.
[41]	Inconsistent system firmware version in parallel mode	Yes	The program (firmware) versions in the parallel system do not match. Parallel operation and AC output stop.
[42]	The parallel communication cable is faulty	Yes	The parallel communication cable is faulty. Parallel operation and AC output stop.
[43]	Serial number error	Yes	The serial number was not set at the factory, or the same factory serial number is used more than once in the system.
[49]	BMS communication error	No	Check that the communication cable is connected correctly. Check that BMS is set to the communication protocol that matches your lithium battery.
[50]	BMS other alarm	No	Check the BMS fault type and troubleshoot the lithium battery problem.
[51]	BMS battery over temperature alarm	No	The BMS reports that the lithium battery is too hot.
[52]	BMS battery over current alarm	No	The BMS reports lithium battery overcurrent.
[53]	BMS battery over voltage alarm	No	The BMS reports lithium battery overvoltage.
[54]	BMS battery low voltage alarm	No	The BMS reports lithium battery low voltage.
[55]	BMS battery low temperature alarm	No	The BMS reports that the lithium battery is too cold.
[73]	DC AFCI Abnormal	No	PV arc fault detected. Check the installation and wiring.
[77]	Grid Circuit Open	No	Check if the AC circuit breaker is closed (on). If it is open, close the breaker and restart the inverter to clear the fault.

9.2 Troubleshooting

Fault code	Fault	What to do
/	No display on the screen	Check that the battery circuit breaker or the PV circuit breaker is closed, and that the switch is in the "ON" position. Press any button on the screen to wake the screen from sleep mode.
[06]	Rechargeable battery overvoltage protection	Measure the battery voltage to see if it is higher than rated. If it is, discharge the battery until its voltage is below the battery's overvoltage recovery point.
[01] [02]	Battery under voltage protection	Charge the battery until it reaches the low voltage disconnection recovery voltage.
[27]	Fan failure	Check if the fan is not turning or is blocked by a foreign object.
[22] [23]	Heat sink over temperature protection	When the unit cools below the recovery temperature, normal charge and discharge control starts again.
[09] [10]	Bypass overload protection, inverter overload protection	1. Use fewer electrical loads. 2. Restart the unit to bring back the load output.
[11]	Inverter short circuit protection	1. Check the load wiring carefully and fix the short-circuit points. 2. Power the unit up again to bring back the load output.
[18]	PV overvoltage	Use a multimeter to check if the PV input voltage is higher than the maximum rated input voltage.
[05]	Battery missed alarm	Check if the battery is not connected, or if the battery circuit breaker is not closed.
[40] [42]	Parallel connection fault	Check whether the parallel cable is connected poorly, such as a loose or wrong connection.
[37]	Parallel ID setting error	Check whether the same parallel ID number is set on more than one unit.
[36]	Parallel current sharing fault	Check whether the parallel current sharing cable is connected poorly, such as a loose or wrong connection.
[39]	Inconsistent AC input source in parallel mode	Check whether the parallel AC inputs all come from the same input source. In three-phase mode, if two of the three phases report that the grid (mains) input sources do not match, try swapping the grid input wiring of any two machines. This checks whether the grid input phase sequence does not match the set phase sequence.
[41]	Inconsistent system firmware version in parallel mode	Check whether every inverter has the same software version.

CAUTION If a fault cannot be fixed with the steps in the table above, contact KOOR support at 574-218-1973 for technical help. Do not take the equipment apart yourself.

10. Protection and Maintenance

10.1 Built-in Protections

No.	Protection	Description
1	PV current/power limiting protection	If the charging current or power of the PV array is higher than the PV rating, the unit charges at the rated value.
2	PV night reverse-current protection	At night, the battery voltage is higher than the PV module voltage. This protection keeps the battery from discharging through the PV module.
3	Grid (mains) input over voltage protection	When the grid voltage is higher than 140V, grid charging stops and the unit switches to inverter mode.
4	Grid (mains) input under voltage protection	When the grid voltage is 85V or lower, grid charging stops and the unit switches to inverter mode.
5	Battery over voltage protection	When the battery voltage reaches the overvoltage disconnection point, PV and grid charging of the battery stop automatically. This keeps the battery from being overcharged and damaged.
6	Battery low voltage protection	When the battery voltage reaches the low voltage disconnection point, battery discharging stops automatically. This keeps the battery from being over-discharged and damaged.
7	Load output short circuit protection	If a short circuit at the load output terminal lasts more than 200 milliseconds, the AC output turns off at once.
8	Heat sink over temperature protection	When the internal temperature is too high, the all-in-one unit stops charging and discharging. When the temperature is back to normal, charging and discharging start again.
9	Overload protection	After an overload protection, the output turns back on after 3 minutes. After 5 overload protections in a row, the output stays off until the unit is powered up again. For the exact overload levels and times, see the technical parameters table in this manual.
10	PV reverse polarity protection	If the PV polarity is reversed, the unit will not be damaged.
11	AC reverse protection	Keeps the battery inverter's AC current from flowing backward into the bypass input.
12	Bypass over current protection	Built-in AC input overcurrent protection circuit breaker.
13	Battery input over current protection	When the battery discharge output current is higher than the maximum value for 1 minute, the AC input is switched to the load.
14	Battery input protection	If the battery is connected backward or the inverter is short-circuited, the battery input fuse inside the inverter blows. This keeps the battery from being damaged or causing a fire.
15	Charge short protection	If the external battery port is short-circuited while charging from PV or AC, the inverter protects itself and stops the output current.
16	CAN communication loss protection	In parallel operation, an alarm is given when CAN communication is lost.
17	Parallel connection error protection	In parallel operation, the equipment is protected when the parallel cable is lost (disconnected).
18	Parallel battery voltage difference protection	In parallel operation, the equipment is protected when the battery connections do not match and the battery voltage is very different from what the host detects.

No.	Protection	Description
19	Parallel AC voltage difference protection	In parallel operation, the equipment is protected when the AC IN input connections do not match.
20	Parallel current sharing fault protection	In parallel operation, running equipment is protected when the load on each inverter is very different. This can be caused by a poorly connected current sharing cable or a damaged device.
21	Synchronization signal fault protection	The equipment is protected when there is a fault in the guidance (sync) signal between the parallel buses that makes the inverters act differently.

10.2 Maintenance

To keep the unit working well for a long time, we recommend doing the checks below twice a year.

- Make sure nothing blocks the airflow around the unit. Remove any dirt or debris from the heat sink.
- Check all exposed wires for damage from sunlight, rubbing against nearby objects, drying out, or chewing by insects or rodents. Repair or replace wires if needed.
- Check that the indicators and display match how the device is operating. Watch for any faults or errors on the display, and fix the cause if needed.
- Check all wiring terminals for corrosion, damaged insulation, signs of high heat, or burning/discoloration. Tighten the screws.
- Check for dirt, insect nests, and corrosion. Clean as needed.
- If the surge arrester has failed, replace it right away. This prevents lightning damage to the unit and to other equipment you own.

KOOR is not liable for damage caused by:

1. Improper use, or use in an unsuitable location.
2. A PV module open-circuit voltage higher than the maximum allowed rated voltage.
3. An operating temperature outside the allowed operating temperature range.
4. Taking apart or repairing the all-in-one solar charge inverter without permission.
5. Force majeure (events outside anyone's control): damage that happens while the all-in-one solar charge inverter is being shipped or handled.

WARNING **Danger of electric shock! Before you do any of the work above, disconnect every power source to the all-in-one unit and make sure all capacitors have discharged. Only then check or work on the unit.**

11. Technical Parameters

Model	KR-L12K-SP-BP
Parallel mode	
Permitted parallel number	1~6
AC mode	
Rated input voltage	120(L1/N, L2/N)/240Vac(L1/L2)
Input phase voltage range	(85Vac~140Vac)±2%
Frequency	50Hz/60Hz (auto detection)
Frequency range	47±0.3Hz~55±0.3Hz (50Hz); 57±0.3Hz~65±0.3Hz (60Hz)
Overload/short circuit protection	Circuit breaker
Efficiency	>95%
Conversion (transfer) time (bypass and inverter)	10ms (typical)
AC reverse protection	Yes
Maximum bypass phase overload current	2-pole, 200A/200A (L1/L2)
Inverter mode	
Output voltage waveform	Pure sine wave
Rated output power (VA)	Split phase: 12000
	Single phase: 7200 (limited by the wiring terminals)
Rated output power (W)	Split phase: 12000
	Single phase: 7200 (limited by the wiring terminals)
Power factor	1
Rated output voltage (Vac)	120Vac (L1/N, L2/N)/240Vac (L1/L2)
Output voltage error	±5%
Output frequency range (Hz)	50Hz±0.3Hz/60Hz±0.3Hz
Maximum efficiency	>93%
Overload protection	(105% < load < 110%) ±5%: reports an error and turns off the output after 5 minutes; (110% < load < 125%) ±5%: reports an error and turns off the output after 10 seconds; Load > 125% ±5%: reports an error and turns off the output after 5 seconds
Peak power	Split phase: 18000VA
	Single phase: 18000VA
Motor load capability	6HP
Output short circuit protection	Yes (output shuts off after a 1-second continuous short circuit)

Model	KR-L12K-SP-BP
Bypass circuit breaker phase specification	2-pole, 200A/200A (L1/L2)
Rated battery input voltage	51.2V (minimum starting voltage 44V)
Battery voltage range	40.0Vdc~58.6Vdc ±0.6Vdc (under voltage alarm / shutdown voltage / over voltage alarm / over voltage recovery... can be set on the LCD screen)
Power saving mode self-consumption	Enters power-saving mode when the load on each phase is zero or less than 25W
AC charging	
Battery type	Lead acid or lithium battery
Maximum charge current (can be set)	160A
Charge current error	±5Adc
Charge voltage range	40–58.6Vdc
Short circuit protection	Circuit breaker and blown fuse
Circuit breaker phase specifications	2-pole, 63A/63A (L1/L2)
Generator mode	
Rated input voltage	120(L1/N, L2/N)/240Vac(L1/L2)
Input phase voltage range	(85Vac~140Vac)±2%
Frequency range	40±0.3Hz~70±0.3Hz
Maximum charge current (can be set)	160A
Overload/short circuit protection	Circuit breaker
Circuit breaker phase specifications	2-pole, 63A/63A (L1/L2)
PV charging	
Number of MPPT groups	2
Maximum PV open circuit voltage	500Vdc
PV operating voltage range	90Vdc~500Vdc
MPPT voltage range	90Vdc~450Vdc
Battery voltage range	40–58.6Vdc
Maximum PV input power	12000W + 12000W
Maximum PV input current	32A + 32A
PV charge current (can be set)	250A
Charging short circuit protection	Circuit breaker

Model	KR-L12K-SP-BP
Wiring protection	Reverse polarity protection
Hybrid charging: maximum charger current (AC charger + PV charger)	
Maximum charger current (can be set)	0–250A
Certified specifications	
EMC certification level	See product label
Operating temperature range	-10℃ to 55℃ (14℉ to 131℉) (>45℃ (113℉) derating)
Storage temperature range	-25℃ ~ 60℃ (-13℉ ~ 140℉)
Humidity range	5% to 95% (conformal coating protection)
Noise	≤60dB
Heat dissipation	Forced air cooling, variable-speed fan
Communication interface	USB, RS485 (Wi-Fi/GPRS), dry contact control
Size (L*W*D)	760mm*485mm*160.8mm (about 29.92 in × 19.09 in × 6.33 in)
Net weight (kg)	35.5 (about 78.3 lb)

Appendix A. Setting Up a KOOR Battery

This appendix covers the two KOOR lithium batteries: the **KOOR Active Stack** (stackable, 51.2V 314Ah, 16 kWh) and the **KOOR 16K Pro** tower. Both talk to this inverter on the **Pylon** protocol over **CAN**.

> Turn the inverter OFF and switch the battery off before you plug in or unplug a communication cable. Set the address switches with the battery off.

A.1 The Rules That Apply to Both

- **Only one battery talks to the inverter.** That one is the master and its address switches are **all OFF (address 0)**. Its CAN jack goes to the inverter's battery communication (BMS) port.
- **The rest chain in a line on RS485-2** — out of B on one battery, into A on the next. Never run a cable from the last one back to the master.
- **The cable to the inverter is a plain straight ethernet cable** for both batteries.
- **No two batteries share an address.**
- The inverter's BMS port has **CAN-H on pin 4 and CAN-L on pin 5**. RS485 sits on pins 1 and 2, repeated on pins 7 and 8.

A.2 Address Switches (both batteries)

Address	1	2	3	4
0 — master	off	off	off	off
1	**ON**	off	off	off
2	off	**ON**	off	off
3	**ON**	**ON**	off	off
4	off	off	**ON**	off
5	**ON**	off	**ON**	off
6	off	**ON**	**ON**	off
7	**ON**	**ON**	**ON**	off
8	off	off	off	**ON**
9	**ON**	off	off	**ON**
10	off	**ON**	off	**ON**
11	**ON**	**ON**	off	**ON**
12	off	off	**ON**	**ON**
13	**ON**	off	**ON**	**ON**
14	off	**ON**	**ON**	**ON**

Up is ON. Set the switches with the battery switched off. Up to 15 batteries: the master is 0, the rest 1 to 14. Changed a switch with the battery on? Hold **RST 3 s** (sleep), then **3 s again** (wake) so it reads the new address.

A.3 KOOR Active Stack

Go by the words stamped on the battery panel. The pair marked **RS485-1 / CAN** is the inverter pair: **CAN** (the right-hand jack) goes to the inverter, RS485-1 is the backup path. The dual jack marked **RS485-2** (A and B) is for chaining pack to pack. RS232 (RJ11) is for the battery software only.

Jack	What it is	Where it goes
CAN	CAN	Master only → inverter BMS port
RS485-1	RS485	Backup path to the inverter (see A.5)
RS485-2 — A and B	RS485	Pack to pack: out of B, into A on the next pack
RS232 (RJ11)	Service	Battery software only

Cable: a plain straight ethernet cable (CAT5e or CAT6), both ends wired the same. Pin 4 is CAN-H and pin 5 is CAN-L, which lines up with the inverter. The master's RS485-2 A jack and the last pack's B jack stay empty. RS485 runs at 9600 baud.

A.4 KOOR 16K Pro

The 16K Pro tower wires **exactly like the Active Stack**. Its panel reads ON/OFF, RST, ADD (the address switches), DRY, then the jacks **RS485-1**, **CAN**, **RS232** and **RS485-2** (a dual jack, A and B).

Jack	Where it goes
CAN	Master tower only → inverter BMS port. Every other tower's CAN jack stays empty.
RS485-2 — A and B	Tower to tower: out of B, into A on the next tower. Master A and last tower B stay empty.
RS485-1	Backup path to the inverter only. Empty on a normal job.
RS232 and DRY	Not used. A chain cable never goes in these.

> **Same cable as the Active Stack — a plain straight ethernet cable.** Nothing crossed, nothing custom, nothing to crimp.

A.5 Inverter Settings

Set these in order, then save.

Setting	Set it to	Why
Battery type	LFP16	16 cells in series, 51.2V nominal.
BMS Setting	BMS-CAN	Turns on BMS communication over CAN.
BMS Protocol Select	Pylon	Both KOOR batteries speak Pylon.
Charge Logic Select	BMS	Lets the battery BMS set the charge current limit.

> **Backup path:** if there is still no battery data, you can try RS485 instead. Move the cable to the battery's **RS485-1** jack (a plain straight cable works there too), then set **BMS Setting** to **BMS-485** and **BMS Protocol Select** to **Pylon**.

> **Set Pylon on the battery too — not just the inverter.** Both ends must be on it, and some packs come set to a different protocol. On every pack set **CAN protocol = Pylon** (Active Stack: battery Bluetooth app, Settings, password **1234**), then hold RST 3 s, then 3 s again.

A.6 How to Tell It Is Talking

- The screen shows the battery **state of charge as a percent** from the BMS, with battery voltage and current.
- No BMS fault code is showing.
- Charge current follows the battery's own limit instead of the value you set by hand.

A.7 If It Will Not Talk

What you see	What to check, in this order
No battery data at all	Is the battery on Pylon? Set CAN protocol = Pylon on every pack and restart it. Then the cable — a plain straight ethernet cable in the CAN jack, not RS485-1 or RS232; swap in a known-good one. Then the master's address (all four switches OFF). Then the inverter protocol setting.
BMS communication fault code	Reseat the RJ45 at both ends until it clicks. Battery switched on with its breaker closed. Only one battery cabled to the inverter.
Data shows, then drops out	Two batteries on the same address, or the chain loops back to the master.
Charging stops early	Normal — the BMS is limiting charge. Check battery temperature and the charge stop settings.

Still stuck? Call KOOR support at 574-218-1973 with the inverter model, the battery model, and the fault code on the screen.